

Generative Modeling for Diffusions with Applications in Machine Learning

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Abstract

Conditioned diffusion processes can be simulated using guided proposals when the transition density of the original diffusion is intractable. However, replacing the true conditioning function with one obtained from a tractable auxiliary process introduces a discrepancy between the guided and conditioned laws. We study when the change of measure relationship between these laws, valid for $t < T$, extends to the terminal time T . Localisation and Lyapunov methods are used to control the singular behaviour arising near the terminal time, and the resulting conditions are examined in a one-dimensional diffusion setting. We then connect this framework to neural guided diffusion bridges and, through a simple Brownian bridge example, derive the variational loss directly and recover the optimal neural correction.

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Chapter 1

Introduction

Diffusion processes are continuous time stochastic models used to describe systems that evolve randomly over time, with applications in areas such as mathematical finance, biology, and machine learning. In many applications, we are interested in what happens when we know something about the process at a later time, such as its value at the end of the time interval. For example, instead of studying the original diffusion process X , one may be interested in its law conditioned on $X_T = v$ where v is the observed terminal position of our process. Conditioning a diffusion on such a terminal observation gives rise to a diffusion bridge.

A diffusion bridge can be constructed theoretically using Doob's h-transform, the resulting drift depends on the transition density of the original diffusion process which is generally intractable for nonlinear models making direct simulations of the conditioned process difficult. Guided Proposals address this issue by replacing the intractable conditioning term with one derived from a tractable auxiliary process. The resulting process can be simulated, but does not in general have the same law as the true conditioned diffusion.

The simulation of conditioned diffusion processes has been studied through a number of different approaches. An early construction was proposed by Clark (1990), who considered one-dimensional diffusions conditioned on a fixed terminal value and introduced an additional Brownian bridge drift to guide the process towards its prescribed terminal value. Related simulation based methods were developed in the context of inference for discretely observed diffusions. For example, Pedersen (1995) proposed a simulation based approximation of otherwise intractable transition densities, while Roberts and Stramer (2001) considered the use of latent diffusion paths for inference

in partially observed nonlinear diffusion models. Durham and Gallant (2002) developed numerical techniques to improve the computational efficiency of such likelihood based methods. Delyon and Hu (2006) considered the simulation of diffusions conditioned on a terminal point more directly, constructing tractable proposal processes and proving that the conditioned process is absolutely continuous with respect to the proposal process, with an expression for the corresponding Radon-Nikodym derivative.

The guided proposal framework used in this dissertation was developed by Schauer et al. (2017). Guided proposals form a particular class of proposal processes, constructed by replacing the intractable conditioning term with one obtained from an auxiliary diffusion with known transition densities. Bierkens et al. (2020) subsequently extended this framework to both elliptic and hypo-elliptic diffusions and to partial terminal observations. Corstanje et al. (2023) extended the guiding framework to continuous time Markov processes and studied conditions under which the guided and conditioned laws remain related at the terminal time.

An important question is therefore how the law of the guided process is related to that of the true conditioned process. Under suitable assumptions, the guided and conditioned laws are equivalent on every interval $[0, t]$, with $t < T$, and are related through a Radon-Nikodym derivative.

However, as we approach the terminal time T , the situation becomes more delicate as the equivalence of the processes on the interval $[0, T)$ for $t < T$ does not automatically imply equivalence on the whole interval $[0, T]$. The difficulty arises from the singular behaviour of the conditioning as $t \uparrow T$: the conditioning function becomes more and more concentrated around the terminal observation and quantities entering the likelihood ratio may become unbounded. Therefore, additional control is required to pass to the terminal limit.

Diffusion processes have also become important in machine learning, particularly in generative modelling. Diffusion probabilistic models generate samples through a gradual noising process together with a learned reverse transformation (Sohl-Dickstein et al., 2015; Ho et al., 2020), while Song et al. (2021) formulate score based generative modelling in continuous time using stochastic differential equations. These ideas have also motivated new approaches specifically for diffusion bridges. Heng et al. (2025) develop a score matching approach based on time reversal, while Baker et al. (2025) propose learning the bridge score directly without first learning a time reversed process. More recently, Yang et al. (2025) construct neural guided

diffusion bridges by augmenting a classical guided proposal with a learnable correction term.

The contribution of this dissertation lies in providing a clearer and more tractable treatment of the general theory of guided diffusion bridges through a detailed specialisation of the existing framework. We present the terminal time proof of Corstanje et al. (2023) in a form that allows the role of each assumption, the localisation procedure and the Lyapunov method to be followed separately, and then specialise their general diffusion setting to a one-dimensional, fully observed model in which the relevant quantities can be derived and analysed directly. This simplified setting is used not only to verify the terminal time conditions, but also to study the choice of Lyapunov function by comparing the logarithmic Lyapunov function with an alternative polynomial choice. Finally, we construct a simple neural guided diffusion bridge example and derive the loss function as a function of θ , and verify that the minimiser recovers the correction required to transform the guided proposal into the true bridge.

The remainder of the dissertation is organised as follows. Chapter 2 introduces the necessary probabilistic background, and Chapter 3 develops diffusion bridges, guided proposals and their finite time change of measure relationship. Chapter 4 studies the extension of this relationship to the terminal time in a general Markov process setting. Chapter 5 specialises the theory to a one-dimensional conditioned diffusion, focusing on the specific choice of the Lyapunov function, and connects the classical framework to neural guided diffusion bridges. Chapter 6 concludes with a discussion of the results, limitations, and possible extensions.

Chapter 2

Preliminaries

This chapter introduces the probabilistic notation and results required to study conditioned diffusions and guided proposals. Transition densities and generators are used to construct and characterise conditioned processes, while absolute continuity, Radon-Nikodym derivatives and Girsanov's theorem provide the change of measure tools needed to compare their probability laws. Finally, Doob's h-transform gives the theoretical construction of the conditioned diffusion considered in Chapter 3.

2.1 Stochastic Differential Equations and Notation

Let $T > 0$ be the terminal time. Throughout the dissertation, we work on the probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$ with $W = (W_t)_{0 \leq t \leq T}$ to be a d' -dimensional standard Brownian motion.

We consider an $\mathbb{R}^d$ -valued diffusion process $(X_t)_{0 \leq t \leq T}$ satisfying the SDE

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t$$

with initial condition $X_0 = x_0$, where $b : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the drift coefficient and $\sigma : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d'}$ is the diffusion coefficient. We write $a(t, x) = \sigma(t, x)\sigma(t, x)^\top$ for the associated diffusion matrix.

Unless stated otherwise, we will assume that the coefficients b and σ satisfy regularity conditions ensuring existence and uniqueness of a solution to the stochastic differential equation. A standard sufficient set of assumptions is

that b and σ are Lipschitz continuous with respect to x and satisfy a linear growth condition; see, for example, Øksendal (2003).

Throughout the dissertation, we use distinct notation for the different processes involved in the construction of diffusion bridges:

- X denotes the original, unconditioned diffusion,
- X^* for the corresponding diffusion conditioned on a future observation,
- $\tilde{X}$ for an auxiliary diffusion chosen to have tractable transition densities,
- X° for the guided process constructed using this auxiliary process.

Their precise definitions and the relationships between these processes will be introduced in Chapters 2 and 3.

We denote by $\mathbb{P}$, $\mathbb{P}^*$, and $\mathbb{P}^\circ$ the probability measures, or equivalently the laws, associated with the original, conditioned, and guided processes, respectively. The probability measures restricted to the filtration $\mathcal{F}_t$ will be denoted by $\mathbb{P}_t$, $\mathbb{P}_t^*$, and $\mathbb{P}_t^\circ$ respectively.

2.2 Markov Processes, Transition Densities, and Generators

Diffusion processes are Markov processes, meaning that conditional on the present state of the process, its future evolution is independent of the past. Hence, the current state contains all the information required to determine the conditional distribution of future states.

More formally, following Billingsley (1995), for times $0 \leq s < t \leq T$ and for any bounded and measurable f ,

$$\mathbb{E}[f(X_t) \mid \mathcal{F}_s] = \mathbb{E}[f(X_t) \mid X_s].$$

Using the notation in Yang et al. (2025), we denote the transition operator from time s to time t by $\mathbb{P}_{s,t}(f(x)) := \mathbb{E}[f(X_t) \mid X_s = x]$. If the process admits a transition density with respect to the Lebesgue measure, then

$$\mathbb{P}_{s,t}(f(x)) = \int_{\mathbb{R}^d} f(y)p(s, x; t, y)dy.$$

For any Borel set $A \subseteq \mathbb{R}^d$,

$$\mathbb{P}(X_t \in A \mid X_s = x) = \int_A p(s, x; t, y) dy, \quad 0 \leq s < t \leq T$$

where $p(s, x; t, y)$ defines the transition density of the process being at position y by time t conditional on being at state x at earlier time s .

For the diffusion introduced in Section 2.1, and unless stated otherwise, we refer to Corstanje et al. (2023) for the remaining of this sections definitions:

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t$$

with diffusion matrix $a(t, x) = \sigma(t, x)\sigma(t, x)^\top$, the infinitesimal generator at time t is given, for a sufficiently smooth function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, by

$$\mathcal{L}_t f(x) = b(t, x)^\top \nabla f(x) + \frac{1}{2} \text{tr}(a(t, x) \nabla^2 f(x)).$$

The infinitesimal generator describes the short time behaviour of the diffusion and provides a useful way of studying functions of the process. For a sufficiently smooth space-time function $f(t, x)$, Itô's formula gives:

$$df(t, X_t) = \left(\frac{\partial f}{\partial t}(t, X_t) + \mathcal{L}_t f(t, X_t) \right) dt + \nabla f(t, X_t)^\top \sigma(t, X_t) dW_t$$

where we denote $\mathcal{A} = \partial_t + \mathcal{L}_t$, so $\mathcal{A}f(t, x) = \partial_t f(t, x) + \mathcal{L}_t f(t, x)$, also known as the space-time generator.

Associated with the generator is a very useful martingale property. For a well behaved function f in the domain of $\mathcal{L}$,

$$M_t^f = f(X_t) - f(X_0) - \int_0^t \mathcal{L}_s f(X_s) ds$$

is a local martingale. More generally, for a well behaved space-time function g ,

$$M_t^g = g(t, X_t) - g(0, X_0) - \int_0^t \mathcal{A}g(s, X_s) ds \quad (2.1)$$

is a local martingale with $\mathcal{A} = \partial_t + \mathcal{L}_t$. Note that a well-behaved function h satisfying $\mathcal{A}h(t, x) = 0$ is referred to as space-time harmonic with respect to X .

The transition density is related to the infinitesimal generator through the Kolmogorov backward equation (Schauer et al. (2017)), which says that for

fixed t and y , and under suitable regularity conditions, the transition density $p(s, x; t, y)$ satisfies

$$\frac{\partial}{\partial s} p(s, x; t, y) + \mathcal{L}_s p(s, x; t, y) = 0, \quad s < t.$$

This will be used in Chapter 3 in the construction of diffusion bridges.

2.3 Absolute Continuity and Radon-Nikodym Derivative

We recall the standard notions of absolute continuity and equivalence of measures, together with the Radon-Nikodym theorem from Billingsley (1995).

Definition 2.1. (*Absolute Continuity*)

Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on the probability space. We say that $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$, written $\mathbb{Q} \ll \mathbb{P}$ if $\mathbb{P}(A) = 0 \implies \mathbb{Q}(A) = 0$ for every $A \in \mathcal{F}$.

Absolute continuity allows the two probability measures to assign different probabilities to different events, while preventing $\mathbb{Q}$ from assigning positive probabilities to events that have zero probability under $\mathbb{P}$.

Definition 2.2. If both $\mathbb{P} \ll \mathbb{Q}$ and $\mathbb{Q} \ll \mathbb{P}$, then we say that $\mathbb{P}$ and $\mathbb{Q}$ are equivalent, and we denote it by $\mathbb{P} \sim \mathbb{Q}$. Equivalent measures therefore have the same null sets.

Theorem 2.1. (*Radon-Nikodym Theorem*)

Let $\mathbb{P}$ and $\mathbb{Q}$ be σ -finite measures on the probability space. Assume $\mathbb{P} \ll \mathbb{Q}$. Then, there exists a measurable function $f : \Omega \rightarrow [0, \infty)$ such that

$$1- \mathbb{P}(A) = \int_A f d\mathbb{Q} \quad \forall A \in \mathcal{F}.$$

2-Uniqueness: If g also satisfies the same property, then $f = g$ $\mathbb{Q}$ -almost everywhere.

The function f is called the Radon-Nikodym derivative and it is denoted as:

$$f = \frac{d\mathbb{P}}{d\mathbb{Q}}.$$

The Radon-Nikodym derivative can be interpreted as the weighting factor or likelihood ratio which converts probabilities that are calculated under $\mathbb{Q}$ into probabilities that are calculated under $\mathbb{P}$.

Since we are dealing with stochastic processes, we will be using the probabilistic form of the Radon-Nikodym derivative which we will later be referring to as "the change of measure formula": For any integrable random variable Y ,

$$E_{\mathbb{P}}[Y] = E_{\mathbb{Q}} \left[Y \frac{d\mathbb{P}}{d\mathbb{Q}} \right]. \quad (2.2)$$

When $\mathbb{P} \ll \mathbb{Q}$ and $\mathbb{L} \ll \mathbb{P}$, then by the chain rule we can write $\frac{d\mathbb{L}}{d\mathbb{Q}} = \frac{d\mathbb{L}}{d\mathbb{P}} \frac{d\mathbb{P}}{d\mathbb{Q}}$. Equivalently, when the relevant measures are equivalent, $\frac{d\mathbb{L}}{d\mathbb{P}} = \frac{d\mathbb{L}}{d\mathbb{Q}} \frac{d\mathbb{Q}}{d\mathbb{P}}$. This chain rule will be used later to compare the conditioned and guided measures through their respective densities with respect to the original diffusion measure.

The Radon-Nikodym theorem guarantees the existence of such a density but does not, in general, provide a formula for it. For diffusion processes, an important case occurs when a change of probability measure results in a change to the drift of the process while leaving the diffusion coefficient unchanged, see Øksendal (2003). Girsanov's theorem, introduced in the following section, provides the corresponding Radon-Nikodym derivative in terms of an exponential martingale.

2.4 Girsanov's Theorem

Consider the diffusion that was introduced in Section 2.1,

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t, \quad X_0 = x_0$$

under the probability measure $\mathbb{P}$. Girsanov's theorem, stated in Øksendal (2003) and below, allows us to construct a new probability measure under which the same process has a modified drift. This will be particularly useful when comparing the original, conditioned and guided diffusions, whose dynamics differ through additional drift terms.

Let $u = (u_t)_{0 \leq t \leq T}$ be an $\mathbb{R}^{d'}$ -valued adapted process that satisfies Novikov's condition

$$\mathbb{E}_{\mathbb{P}} \left[\exp \left(\frac{1}{2} \int_0^T \|u_s\|^2 ds \right) \right] < \infty.$$

Define the process

$$M_t = \exp \left(\int_0^t u_s^\top dW_s - \frac{1}{2} \int_0^t u_s^2 ds \right), \quad 0 \leq t \leq T.$$

Under Novikov's condition, $M_t = (M_t)_{0 \leq t \leq T}$ is a martingale, and may be used to define a new probability measure $\mathbb{Q}$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = M_T.$$

Consequently, recalling that $\mathbb{P}_t$ and $\mathbb{Q}_t$ denote the restriction of the measures $\mathbb{P}$ and $\mathbb{Q}$ to the filtration $\mathcal{F}_t$, then

$$\frac{d\mathbb{Q}_t}{d\mathbb{P}_t} = M_t, \quad 0 \leq t \leq T.$$

Girsanov's theorem states that the process $W_t^{\mathbb{Q}} = W_t - \int_0^t u_s ds$ is a Brownian motion under $\mathbb{Q}$. Since $dW_t = dW_t^{\mathbb{Q}} + u_t dt$, we can substitute this into the original SDE:

$$dX_t = (b(t, X_t) + \sigma(t, X_t)u_t)dt + \sigma(t, X_t)dW_t^{\mathbb{Q}}, \quad X_0 = x_0.$$

Therefore, under the new probability measure $\mathbb{Q}$, the drift changes from $b(t, x)$ to $b(t, x) + \sigma(t, x)u_t$ while the diffusion coefficient $\sigma(t, x)$ remains unchanged.

The exponential martingale M_t is precisely the Radon-Nikodym derivative that is introduced in Section 2.3 and therefore provides the likelihood ratio associated with this change of measure.

In the Doob's h-transform considered below, choosing $u_t = \sigma(t, X_t)^{\top} \nabla \log h(t, X_t)$ produces the additional drift $a(t, X_t) \nabla \log h(t, X_t)$.

2.5 Doob's h-transform

We adapt the Doob's h-transform in Bloemendal (2010) to the time inhomogeneous setting considered here. Doob's h-transform provides a way to modify the law of a Markov process through a positive, harmonic function $h : [0, T) \times \mathbb{R}^d \rightarrow (0, \infty)$. In the setting considered here, we take $h(t, x)$ to be a sufficiently regular positive function which is space-time harmonic with respect to the original process X , so that $\mathcal{A}h(t, x) = 0$, allowing us to derive the resulting change in the generator and drift.

For $0 \leq s < t < T$, the h -transformed transition density is defined by:

$$p^h(s, x; t, y) = p(s, x; t, y) \frac{h(t, y)}{h(s, x)}.$$

The denominator normalises the density as

$$h(s, x) = \mathbb{E}[h(t, X_t) \mid X_s = x] = \int_{\mathbb{R}^d} p(s, x; t, y) h(t, y) dy.$$

Hence,

$$\int_{\mathbb{R}^d} p^h(s, x; t, y) dy = \int_{\mathbb{R}^d} \frac{p(s, x; t, y) h(t, y)}{h(s, x)} dy = \frac{\int_{\mathbb{R}^d} p(s, x; t, y) h(t, y) dy}{h(s, x)} = 1.$$

The transformed generator would be:

$$\mathcal{A}^h f = \frac{1}{h} [\mathcal{A}(fh) - f\mathcal{A}h]$$

from Corstanje et al. (2023). Since here $\mathcal{A}h = 0$, then

$$\mathcal{A}^h f = \frac{1}{h} \mathcal{A}(fh).$$

We will expand $\mathcal{A}(fh)$, we know that $\mathcal{A} = \partial_t + \mathcal{L}_t$ from Section 2.2 with $\mathcal{L}_t f = b^\top \nabla f + \frac{1}{2} \text{tr}(a \nabla^2 f)$, $a = \sigma \sigma^\top$. Therefore,

$$\begin{aligned} \mathcal{A}(fh) &= (\partial_t + \mathcal{L}_t)(fh) = \partial_t(fh) + \mathcal{L}_t(fh) \\ &= h\partial_t f + f\partial_t h + b^\top \nabla(fh) + \frac{1}{2} \text{tr}(a \nabla^2(fh)) \end{aligned}$$

where we used the product rule for $\partial_t(fh) = h\partial_t f + f\partial_t h$.

Note that $\nabla(fh) = h\nabla f + f\nabla h$ and $\nabla^2(fh) = h\nabla^2 f + f\nabla^2 h + \nabla f(\nabla h)^\top + \nabla h(\nabla f)^\top$. So,

$$\begin{aligned} \mathcal{A}(fh) &= h\partial_t f + hb^\top \nabla f + \frac{1}{2} \text{tr}(ah\nabla^2 f) + f\partial_t h + fb^\top \nabla h \\ &\quad + \frac{1}{2} \text{tr}(af\nabla^2 h) + \frac{1}{2} \text{tr}(a\nabla f(\nabla h)^\top) + \frac{1}{2} \text{tr}(a\nabla h(\nabla f)^\top) \\ &= h\mathcal{A}f + f\mathcal{A}h + (\nabla f)^\top a \nabla h \end{aligned}$$

we have used the symmetry of $a = \sigma \sigma^\top$ to combine the two cross terms. Hence,

$$\mathcal{A}^h f = \frac{1}{h} [h\mathcal{A}f + f\mathcal{A}h + (\nabla f)^\top a \nabla h]$$

and as $\mathcal{A}h = 0$, then:

$$\begin{aligned} \mathcal{A}^h f &= \frac{1}{h} [h\mathcal{A}f + (\nabla f)^\top a \nabla h] \\ &= \mathcal{A}f + (\nabla f)^\top a \nabla \log h \end{aligned} \tag{2.3}$$

as $\frac{\nabla h}{h} = \nabla \log h$.

By definition of $\mathcal{A}f$, $\mathcal{A}f = \partial_t f + \mathcal{L}_t f$. Thus, (2.3) becomes $\mathcal{A}^h f = \partial_t f + \mathcal{L}_t f + (\nabla f)^\top a \nabla \log h$, we also know by definition of $\mathcal{A}^h f$ that $\mathcal{A}^h f = \partial_t f + \mathcal{L}_t^h f$. Then, $\mathcal{L}_t^h f = \mathcal{L}_t f + (\nabla f)^\top a \nabla \log h$ and the new spatial generator would be

$$\begin{aligned}\mathcal{L}_t^h f &= \mathcal{L}_t f + (\nabla f)^\top a \nabla \log h \\ &= b(t, x)^\top \nabla f(x) + \frac{1}{2} \text{tr}(a(t, x)) \nabla^2 f(x) + (\nabla f)^\top a \nabla \log h(t, x) \\ &= [b + a \nabla \log h]^\top \nabla f + \frac{1}{2} \text{tr}(a(t, x) \nabla^2 f(x))\end{aligned}$$

where we used definition of $\mathcal{L}_t f$ and the fact that $a = \sigma \sigma^\top$ is symmetric for the last step. We can also see that the transformed diffusion has a new drift: $b(t, x) + a(t, x) \nabla \log h(t, x)$ but the diffusion coefficient remained the same.

This calculation characterises the effect of Doob's h-transform on the dynamics of the original diffusion. In particular, the transformation acts through an additional term in the drift determined by the choice of h . In the following chapter, we specialise this construction to diffusion bridges, by choosing h from the transition density associated with a fixed future observation. This will allow us to describe the conditioned diffusion and subsequently introduce guided processes.

Chapter 3

Diffusion Bridges and Guided Processes

This chapter applies the tools introduced in Chapter 2 to the construction of conditioned diffusions. We first describe diffusion bridges through Doob's h-transform and then introduce guided proposals as a tractable alternative when the required transition density is unavailable. Finally, we study the relationship between the conditioned and guided laws on finite time intervals through a change of measure.

3.1 Diffusion Bridges

As mentioned earlier, diffusion bridges are stochastic processes that start at a specific point and are conditioned to end at another specific point. Consider the original diffusion process $dX_t = b(X_t)dt + \sigma(X_t)dW_t$ and suppose that we wish to condition on the terminal observation $X_T = v$.

If we were to immediately compute $p(X_t | X_T = v)$ using Bayes formula, we would obtain:

$$\mathbb{P}(X_t | X_T = v) = \frac{\mathbb{P}(X_T = v | X_t = x)\mathbb{P}(X_t = x | X_0 = x_0)}{\mathbb{P}(X_T = v | X_0 = x_0)}.$$

However, as we are in a continuous state space, typically $\mathbb{P}(X_T = v) = 0$. Thus, conditioning on the event $\{X_T = v\}$ cannot be interpreted using the elementary form of Bayes formula for events of positive probability. Instead, we construct the bridge through the transition density and Doob's h-transform rather than by naively dividing the probabilities.

By setting $h(t, x) = p(t, x; T, v)$, the transition density of the conditioned process, obtained from the Doob's h-transform introduced in Section 2.5, is

$$p^*(s, x; t, y) = p(s, x; t, y) \frac{h(t, y)}{h(s, x)}.$$

By the Kolmogorov backward equation introduced at the end of Section 2.2, fixing the terminal pair to be (T, v) , and as $h(t, x) = p(t, x; T, v)$ we satisfy

$$\frac{\partial}{\partial t} p(t, x; T, v) + \mathcal{L}_t p(t, x; T, v) = \frac{\partial}{\partial t} h(t, x) + \mathcal{L}_t h(t, x) = 0, \quad t < T.$$

Hence, h is space-time harmonic with respect to the space-time generator $\mathcal{A} = \partial_t + \mathcal{L}_t$ so that

$$\mathcal{A}h(t, x) = 0, \quad t < T.$$

As $t \uparrow T$, $h(t, \cdot)$ becomes increasingly concentrated around v , with its terminal behaviour corresponding formally to a Dirac mass at v .

Using Doob's h-transform from Section 2.5 in Chapter 2, we obtain the conditioned process:

$$dX_t^* = (b(t, X_t^*) + a(t, X_t^*) \nabla \log h(t, X_t^*)) dt + \sigma(t, X_t^*) dW_t^*$$

where W^* is Brownian motion under $\mathbb{P}^*$, with initial condition $X_0^* = x_0$, $a(t, X_t^*) = \sigma(t, X_t^*) \sigma^\top(t, X_t^*)$, drift and diffusion terms $b(t, X_t^*) + \sigma \sigma^\top \nabla \log h(t, X_t^*)$, $\sigma(t, X_t^*)$, respectively. Defining $r(t, x) = \nabla \log h(t, x)$, the additional term in the drift $\sigma \sigma^\top r(t, x)$ pulls the process towards the terminal endpoint v .

Therefore, if the transition density of the true diffusion process were known, then we can simulate the diffusion bridge through the modified drift. However, in most nonlinear diffusion bridges, the transition density $p(t, x; T, v)$ and hence $\nabla \log h(t, x)$ is not known, making it unavailable.

3.2 Guided Proposals

Instead of simulating from the conditioned diffusion process itself, we would instead simulate from another process, called the guiding process. The guiding process replaces the unknown term $\nabla \log h(t, x)$ with $\nabla \log \tilde{h}(t, x)$ where $\tilde{h}(t, x)$ is a known transition density as it comes from the simpler auxiliary process $\tilde{X}$. The guided proposal construction considered here follows Schauer et al. (2017).

The auxiliary diffusion $\tilde{X}$ whose transition density $\tilde{p}$ is available in closed form is written as:

$$d\tilde{X}_t = \tilde{b}(t, \tilde{X}_t)dt + \tilde{\sigma}(t, \tilde{X}_t)dW_t, \quad \tilde{X}_0 = x_0$$

with $\tilde{h}(t, x) = \tilde{p}(t, x; T, v)$.

It is important to distinguish the auxiliary process $\tilde{X}$ from the guided process X° . The auxiliary process is introduced only to provide a tractable approximation $\tilde{h}$ to the unknown conditioning function h . The guided process itself retains the original drift and diffusion coefficients, with an additional guiding term constructed from $\tilde{h}$.

Thus, the guided process is:

$$dX_t^\circ = [b(t, X_t^\circ) + a(t, X_t^\circ)\tilde{r}(t, X_t^\circ)]dt + \sigma(t, X_t^\circ)dW_t^\circ, \quad X_0 = x_0$$

where $a(t, X_t^\circ) = \sigma\sigma^\top$ and $\tilde{r}(t, x) := \nabla \log \tilde{h}(t, x)$.

As we have defined $r(t, x)$ and $\tilde{r}(t, x)$, the drifts of the conditioned and guided processes are: $b^*(t, x) = b(t, x) + a(t, x)r(t, x)$ and $b^\circ(t, x) = b(t, x) + a(t, x)\tilde{r}(t, x)$, respectively. Therefore,

$$\begin{aligned} b^*(t, x) - b^\circ(t, x) &= a(t, x)r(t, x) - a(t, x)\tilde{r}(t, x) \\ &= a(t, x)[r(t, x) - \tilde{r}(t, x)]. \end{aligned}$$

Thus, the accuracy of the guided drift depends on how closely $\tilde{r}(t, x) = \nabla \log \tilde{h}(t, x)$ approximates the true score $r(t, x) = \nabla \log h(t, x)$. Their difference determines the discrepancy between the bridge and guided drifts.

The following section makes this relationship precise by deriving the Radon-Nikodym derivative between the two laws on intervals strictly before T .

3.3 Relationship between Bridge and Guided Processes

As the guided process is constructed using the tractable function $\tilde{h}$ rather than the exact conditioning function h , the law $\mathbb{P}^\circ$ does not coincide with the bridge law $\mathbb{P}^*$. We therefore need to determine when the two measures are equivalent, and when they are, to compute the Radon-Nikodym derivative. We first consider the question on a finite interval $[0, t]$ with $t < T$.

To define these changes of measure consistently, we would need the associated density process to be a martingale. Following Corstanje et al. (2023), this motivates the following definition of an S-good function.

Definition 3.1. (*S-good function*)

Let $S \leq T$. We call $h \in D(\mathcal{A})$, where $D(\mathcal{A})$ is the domain of the space-time generator, an S-good function if $h > 0$ and

$$D_t^h = \frac{h(t, X_t)}{h(0, x_0)} \exp \left(- \int_0^t \frac{\mathcal{A}h}{h}(s, X_s) ds \right), \quad 0 \leq t \leq S$$

is a martingale adapted to the filtration $\{\mathcal{F}_t\}_{t \in [0, S]}$.

We can now write the definition from Corstanje et al. (2023) of a conditioned process.

Definition 3.2. (*Conditioned Process*)

Let $S \leq T$ and h an S-good function, we define the change of measure

$$d\mathbb{P}_t^* = D_t^h d\mathbb{P}_t, \quad t \in [0, S]$$

where $\mathbb{P}_t^*$ is the conditioned measure induced by h .

As h is space-time harmonic, $\mathcal{A}h = 0$ and the density simplifies to

$$D_t^h = \frac{h(t, X_t)}{h(0, x_0)}.$$

However, for $\tilde{h}$, $\mathcal{A}\tilde{h}$ does not generally vanish as $\tilde{h}$ comes from the auxiliary process rather than the original one. Thus,

$$D_t^{\tilde{h}} = \frac{\tilde{h}(t, X_t)}{\tilde{h}(0, x_0)} \frac{1}{\Psi_t(X)}$$

where $\Psi_t(X) = \exp \left(\int_0^t \frac{\mathcal{A}\tilde{h}}{\tilde{h}}(s, X_s) ds \right)$ which acts as a path dependent correction factor accounting for the discrepancy between the original and auxiliary processes.

As $D_t^{\tilde{h}}$ defines the guided measure, we can define the change of measure, from Corstanje et al. (2023), as

$$d\mathbb{P}_t^\circ = D_t^{\tilde{h}} d\mathbb{P}_t. \tag{3.1}$$

It is important to note that when $\tilde{h}$ is harmonic under the auxiliary generator $\tilde{\mathcal{A}}$ i.e. $\tilde{\mathcal{A}}\tilde{h} = 0$. Then $\mathcal{A}\tilde{h} = (\mathcal{A} - \tilde{\mathcal{A}})\tilde{h}$ and the correction factor becomes

$$\Psi_t(X) = \exp \left(\int_0^t \frac{(\mathcal{A} - \tilde{\mathcal{A}})\tilde{h}}{\tilde{h}}(s, X_s) ds \right).$$

The proposition below follows from Corstanje et al. (2023). However, we develop the proof further to show the equivalence and each step in the derivation of the Radon-Nikodym derivative.

Proposition 3.1. *Suppose that h and $\tilde{h}$ are S -good functions for a fixed $S < T$ and that h is space-time harmonic for $\mathcal{A}$ i.e. $\mathcal{A}h = 0$. Assume $D_t^{\tilde{h}} > 0$ on $[0, S]$. Then, for all $t \leq S$,*

$$\mathbb{P}_t^* \sim \mathbb{P}_t^\circ$$

and

$$\frac{d\mathbb{P}_t^*}{d\mathbb{P}_t^\circ} = \frac{h(t, X_t)}{\tilde{h}(t, X_t)} \frac{\tilde{h}(0, x_0)}{h(0, x_0)} \Psi_t(X), \quad t \in [0, S]$$

where $\Psi_t(X) = \exp(\int_0^t \frac{\mathcal{A}\tilde{h}}{\tilde{h}}(s, X_s) ds)$.

Proof: As h is an S -good function then $h > 0$, $\tilde{h} > 0$, and $D_t^{\tilde{h}} > 0$ by assumption, then both $\mathbb{P}_t^*$ and $\mathbb{P}_t^\circ$ are equivalent to $\mathbb{P}_t$.

In order to see how the equivalence follows, we will start by showing $\mathbb{P}_t^\circ \ll \mathbb{P}_t$. Consider $B \in \mathcal{F}_t$ so that $\mathbb{P}_t(B) = 0$. By solving the SDE in (3.1), we get

$$\mathbb{P}_t^\circ(B) = \int_B D_t^{\tilde{h}} d\mathbb{P}_t = 0.$$

As $D_t^{\tilde{h}} > 0$, then $\mathbb{P}_t(B) = 0$ and $\mathbb{P}_t^\circ \ll \mathbb{P}_t$. Moreover, as $D_t^{\tilde{h}} > 0$ almost surely, its reciprocal is well defined and $\frac{d\mathbb{P}_t}{d\mathbb{P}_t^\circ} = \frac{1}{D_t^{\tilde{h}}}$ which similarly gives $\mathbb{P}_t \ll \mathbb{P}_t^\circ$.

Therefore, $\mathbb{P}_t \sim \mathbb{P}_t^\circ$.

Using the same reasoning, we can also obtain the other equivalence $\mathbb{P}_t \sim \mathbb{P}_t^*$ using the Doob's h-transform considered in Definition 3.2 ($d\mathbb{P}_t^* = D_t^h d\mathbb{P}_t$).

Going back to the proof, we now know that

$$d\mathbb{P}_t = (D_t^{\tilde{h}})^{-1} d\mathbb{P}_t^\circ$$

and thus

$$\begin{aligned}
\frac{d\mathbb{P}_t^*}{d\mathbb{P}_t^\circ} &= \frac{d\mathbb{P}_t^*}{d\mathbb{P}_t} \frac{d\mathbb{P}_t}{d\mathbb{P}_t^\circ} \\
&= \frac{D_t^h}{D_t^{\tilde{h}}} \\
&= \frac{h(t, X_t)}{h(0, x_0)} \exp \left(- \int_0^t \frac{\mathcal{A}h}{h}(s, X_s) ds \right) \times \frac{\tilde{h}(0, x_0)}{\tilde{h}(t, X_t)} \frac{1}{\exp(- \int_0^t \frac{\mathcal{A}\tilde{h}}{\tilde{h}}(s, X_s) ds)} \\
&\stackrel{\mathcal{A}h=0}{=} \frac{h(t, X_t)}{h(0, x_0)} \frac{\tilde{h}(0, x_0)}{\tilde{h}(t, X_t)} \exp \left(\int_0^t \frac{\mathcal{A}\tilde{h}}{\tilde{h}}(s, X_s) ds \right) = \frac{h(t, X_t)}{h(0, x_0)} \frac{\tilde{h}(0, x_0)}{\tilde{h}(t, X_t)} \Psi_t(X). \blacksquare
\end{aligned}$$

Proposition 3.1 establishes equivalence between the bridge and guided laws on every finite interval $[0, t]$ with $t < T$. This does not, however, immediately imply equivalence on the full interval $[0, T]$. As $t \uparrow T$, the conditioning becomes increasingly singular near the prescribed endpoint, while the terms appearing in the Radon-Nikodym derivative may not remain bounded or converge. Additional assumptions are therefore required to control the terminal limit. These conditions and the corresponding extension of the change of measure formula to T are addressed in Chapter 4.

Chapter 4

Change of Measure at the Terminal Time

This chapter develops the terminal time extension in four stages. We first introduce and interpret the assumptions required for the limiting procedure. We then construct suitable localising events using a Lyapunov function, define the terminal correction factor, and finally establish the change of measure result at time T .

4.1 Assumptions for Terminal Time Extension

We follow the terminal time framework of Corstanje et al. (2023). Their main contribution is to extend the guided process construction beyond diffusion processes to a general class of continuous time Markov processes and to provide sufficient conditions under which the finite time change of measure relationship extends to the terminal time. In particular, Theorem 3.3 of Corstanje et al. (2023) gives the terminal time result on which Theorem 4.1 below is based.

A consequence of this generality is that the sufficient conditions are abstract, making the role of the assumptions and the localisation procedure difficult to see directly. Here, we separate the roles of the assumptions, and develop the localisation, Lyapunov, and terminal time arguments more fully. This also prepares for the one-dimensional diffusion specialisation in Chapter 5,

where these conditions can be examined in more detail.

We will recall and introduce notations used for this chapter:

- h : the exact conditioning function, usually involving the unknown transition density.
- $\tilde{h}$: the tractable approximation used to construct the guiding process.
- $\Psi_t(X)$: the correction factor accounting for the difference between the true and auxiliary process.
- $\kappa(t)$: a deterministic scaling factor (corrects a possible difference in terminal behaviour between h and $\tilde{h}$).

We introduce a family of events $A_k(t)$, indexed by $k \in \mathbb{N}$ and $t < T$, which will be used to localise the passage to the terminal time to paths on which the relevant quantities remain controlled. The index k represents the level of localisation: larger values of k correspond to less restrictive events and therefore include more sample paths. At this stage we do not yet specify the construction of $A_k(t)$. In Section 4.2, these events will be constructed using a non-negative Lyapunov function V , with $A_k(t)$ consisting of paths for which V remains below the level k up to time t .

The following assumptions are adapted from the terminal time conditions of Corstanje et al. (2023). We state them separately and provide an interpretation of each condition to clarify how it is used in the proof.

Assumption 4.1. *(The good path events must be nested)*

$A_k(t) \subseteq A_{k+1}(t)$ and $A_k(t) \subseteq A_k(s)$ when $s \leq t$.

The first inclusion means that increasing k gives a less restrictive event so that progressively more sample path are included. The second inclusion means that for a fixed k , the event becomes more restrictive as t increases. Thus, the events $A_k(t)$ form a nested localisation in both k and t .

Assumption 4.2. *(The scaled guide must recover the true conditioning function)*

The original process X has the transition density p under $\mathbb{P}$ with respect to the dominating measure ν , that is

$$\mathbb{P}(X_t \in A | X_s = x) = \int_A p(s, x; t, y) d\nu(y).$$

Define $\hat{h}(t; s, x) = \int \kappa(t) \tilde{h}(t, y) p(s, x; t, y) \nu(dy)$ and we require $\lim_{t \uparrow T} \hat{h}(t; s, x) = h(s, x)$.

To better interpret what $\hat{h}(t; s, x)$, one can think of it as taking the approximating conditioning function $\tilde{h}$ at a time very close to T , scale it by $\kappa(t)$, and average it over where the true process could be at that time, as t approaches T , we should be able to recover the true conditioning function h . Thus,

$$\hat{h}(t; s, x) = \mathbb{E}[\kappa(t) \tilde{h}(t, X_t) | X_s = x].$$

The approximation $\tilde{h}$ does not have to be equal to h pointwise, it is the averaged and scaled version that is needed to recover h .

Assumption 4.3. *(The backward average must not blow up)*
For $s < t < T$, the random variable $\hat{h}(t; s, X_s)$ must be bounded almost surely under the original law $\mathbb{P}$.

Note that Assumption 4.2 gives the required pointwise limiting behaviour whereas Assumption 4.3 ensures that this convergence remains controlled when the fixed state x is replaced by the random state X_s . This condition will be used later when passing to the terminal limit inside an expectation.

Assumption 4.4. *(No probability mass is lost in the limit)*

$$\lim_{k \rightarrow \infty} \lim_{t \uparrow T} \mathbb{E}_t^* \left[\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A_k(t)} \right] = \mathbb{P}_T^*(A(T))$$

where

$$A_k(T) = \bigcap_{t < T} A_k(t), \quad A(t) = \bigcup_k A_k(t), \quad A(T) = \bigcap_{t < T} A(t).$$

The factor $\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t)$ measures the discrepancy between the scaled auxiliary conditioning function and the exact conditioning function. Assumption 4.4 requires the limiting weighted expectation over the localising events to coincide with the probability of the terminal event $A(T)$ under the conditioned law. In this sense, the limiting procedure does not lose probability mass. As the events $A(t)$ become more restrictive as $t \uparrow T$, the event $A(T) = \bigcap_{t < T} A(t)$ can be interpreted as the event that the path continues to satisfy the localisation conditions arbitrarily close to the terminal time.

Assumption 4.5. *(The correction factor must be uniformly bounded on good paths)*

For each fixed k ,

$$\sup_{0 \leq t < T} |\Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)}| < \infty,$$

$\mathbb{P}^\circ$ -almost surely.

In other words, for each k , the quantity $\Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)}$ is uniformly bounded in t , $\mathbb{P}^\circ$ -almost surely. Moreover, for each fixed k ,

$$\lim_{t \uparrow T} \Psi_t(X)\kappa(t)\mathbf{1}_{A_k(T)}$$

exists $\mathbb{P}^\circ$ -almost surely.

From Chapter 3, we saw that when $\tilde{h}$ comes from an auxiliary process, $\Psi_t(X) = \exp(\int_0^t \frac{(A-\tilde{A})\tilde{h}}{\tilde{h}}(s, X_s)ds)$ which may be unbounded. This assumption requires that, after we include the scaling factor $\kappa(t)$, the growth of $\Psi_t(X)$ remains controlled on each localising event $A_k(t)$. The boundedness condition is imposed only locally as global boundedness may be too restrictive because the conditioning becomes singular as $t \uparrow T$.

We additionally require the localised terminal limit to exist. In Corstanje et al. (2023, Lemma 3.2), this limit is introduced from continuity and boundedness of $t \mapsto \Psi_t(X)\kappa(t)$ on $A_k(T)$. Since these properties alone do not guarantee convergence as $t \uparrow T$, we include the existence of the limit as a separate condition here. This ensures that the terminal correction factor used in Lemma 4.1 is well defined.

The scaling factor κ is included to compensate for a possible difference in the terminal scaling of the true and auxiliary conditioning functions. For example, if the auxiliary conditioning function has a deterministic time dependent, normalising factor $c(t)$, one may choose $\kappa(t) = \frac{1}{c(t)}$ to remove this factor and recover the terminal behaviour. Therefore, $\kappa(t)$ corrects the mismatch of the auxiliary process near T . For the diffusion setting considered in Chapter 5, we do not need to rescale allowing us to take $\kappa(t) = 1$.

Assumptions 4.1-4.5 play complementary roles in the terminal time construction. Assumptions 4.2 and 4.3 control the limiting behaviour of the scaled auxiliary conditioning function, Assumption 4.4 prevents loss of mass on the localising events, and Assumption 4.5 controls the correction factor. Assumption 4.1 provides the nested localisation structure required to combine these limits.

4.2 Localisation and Lyapunov Control

The assumptions in the previous sections specify the properties required of the localising events $A_k(t)$, but do not describe how these events should be constructed. A natural way to construct such events is to introduce a non-negative function $V : [t_0, T) \times S \rightarrow [0, \infty)$, and define

$$A_k(t) = \left\{ \sup_{t_0 \leq u < t} V(u, X_u) \leq k \right\}.$$

The function V acts as a control function for the trajectory where a large value of $V(t, X_t)$ indicates that the path has entered a region in which the quantities become difficult to control. The event $A_k(t)$ therefore contains paths for which this control quantity remains below level k up to time t .

At this general stage, V does not need to have one fixed interpretation. Its exact meaning depends on the application, but it should be chosen so that the boundedness of V implies boundedness of the correction factor appearing in Assumption 4.5.

The following proposition is stated by Corstanje et al. (2023, Proposition 3.4), but no proof is provided there. We give a proof here to show how the Lyapunov function is used to define localising events with the required nesting properties and, under almost sure boundedness of the Lyapunov function, ensures that the terminal localisation event $A(T)$ has full probability under the guided law.

Proposition 4.1. *(Construction of the localising events)*

Let $V : [t_0, T) \times S \rightarrow [0, \infty)$ be non-negative and define

$$A_k(t) = \left\{ \sup_{t_0 \leq u < t} V(u, X_u) \leq k \right\}.$$

Then the events satisfy $A_k(t) \subseteq A_{k+1}(t)$ and for $t_0 \leq s \leq t < T$,

$A_k(t) \subseteq A_k(s)$. Consequently, Assumption 4.1 is satisfied.

Moreover, if $\sup_{t_0 \leq t < T} V(t, X_t) < \infty$ $\mathbb{P}^\circ$ -almost surely, then $\mathbb{P}_T^\circ(A(T)) = 1$, simplifying the Radon-Nikodym derivative in Theorem 4.1 below.

Proof:

The first inclusion $A_k(t) \subseteq A_{k+1}(t)$ follows immediately by taking $\omega \in A_k(t)$, and by definition of $A_k(t)$,

$$\sup_{t_0 \leq r < t} V(r, X_r(\omega)) \leq k \leq k + 1$$

hence, $\omega \in A_{k+1}(t)$.

Similarly, for $s \leq t$, and taking $\omega \in A_k(t)$, again by definition of $A_k(t)$,

$$\sup_{t_0 \leq r < s} V(r, X_r(\omega)) \leq \sup_{t_0 \leq r < t} V(r, X_r(\omega)) \leq k$$

proving the second inclusion and thus verifying Assumption 4.1.

Suppose $\sup_{t_0 \leq t < T} V(t, X_t)$ is $\mathbb{P}^\circ$ almost surely bounded. Let

$$B = \sup_{t_0 \leq r < T} V(r, X_r) < \infty$$

and by Definition 4.1 below, $\mathbb{P}_T^\circ(B) = 1$.

Definition 4.1. (*Almost sure boundedness*)

Suppose (X_t) , $t \in [0, T]$ is a process. We say that it is almost surely bounded on $[0, T]$ if

$$\mathbb{P}(\sup_{0 \leq t \leq T} |X_t| < \infty) = 1.$$

Meaning, for almost every sample path ω , there exists a finite bound, which may depend on ω such that $|X(\omega)| \leq M(\omega)$.

We show that $B \subseteq A(T)$ which would allow us to conclude $1 = \mathbb{P}_T^\circ(B) \leq \mathbb{P}_T^\circ(A(T))$ as $\mathbb{P}_T^\circ(B) = 1$ by assumption, then $\mathbb{P}_T^\circ(A(T)) = 1$.

Taking any path $\omega \in B$ and by definition of B and almost sure boundedness,

$$C(\omega) := \sup_{t_0 \leq r < T} V(r, X_r(\omega)) < \infty$$

where we choose an integer $k = k(\omega)$ such that $k > C(\omega)$ where k may depend on the path.

Hence, for every $t < T$,

$$\sup_{t_0 \leq r < t} V(r, X_r(\omega)) \leq \sup_{t_0 \leq r < T} V(r, X_r(\omega)) = C(\omega) < k$$

so $\omega \in A_k(t)$, for every $t < T$. Hence,

$$\omega \in \bigcap_{t < T} A_k(t) = A_k(T)$$

as we defined the integer $k = k(\omega)$ which works for every t , and $A_k(T)$ is as defined in Assumption 4.4, it follows that

$$A_k(T) \subseteq \bigcup_{k=1}^{\infty} A_k(T).$$

Therefore, $\omega \in \bigcup_{k=1}^{\infty} A_k(T)$ and thus $B \subseteq \bigcup_{k=1}^{\infty} A_k(T)$. Moreover, for every k ,

$$A_k(T) = \bigcap_{t < T} A_k(t) \subseteq \bigcap_{t < T} A(t) = A(T)$$

since $A_k(t) \subseteq A(t)$ for every $t < T$. Therefore,

$$\bigcup_{k=1}^{\infty} A_k(T) \subseteq A(T).$$

Hence,

$$B \subseteq \bigcup_{k=1}^{\infty} A_k(T) \subseteq A(T).$$

By assumption, $\mathbb{P}_T^{\circ}(B) = 1$ and thus

$$1 = \mathbb{P}_T^{\circ}(B) \leq \mathbb{P}_T^{\circ}\left(\bigcup_{k=1}^{\infty} A_k(T)\right) \leq \mathbb{P}_T^{\circ}(A(T)) \leq 1$$

concluding the proof as $\mathbb{P}_T^{\circ}(\bigcup_{k=1}^{\infty} A_k(T)) = \mathbb{P}_T^{\circ}(A(T)) = 1$. Moreover,

$$\bigcup_{k=1}^{\infty} A_k(T) = A(T)$$

$\mathbb{P}^{\circ}$ -almost surely. This identification will be used in Theorem 4.1 below. $\blacksquare$

Proposition 4.1 reduces the problem to showing that $V(t, X_t)$ remains bounded along almost every trajectory of the guided process. However, it does not specify how this almost sure boundedness should be obtained. The following proposition provides a sufficient condition by imposing an additional requirement on V .

In the time inhomogeneous setting, a non-negative function V is called a Lyapunov function for the guided process if $V \in D(\mathcal{A})$ and satisfies $\mathcal{A}^{\circ}V(t, x) \leq 0$. This condition means that $V(t, X_t)$ has a non-positive infinitesimal drift under the guided process. Although $V(t, X_t)$ will not decrease along each individual path due to random fluctuations, its conditional expected future value does not exceed its current value i.e. it is a supermartingale, $\mathbb{E}^{\circ}[V(t, X_t) \mid \mathcal{F}_s] \leq V(s, X_s)$, $s \leq t$.

The following proposition is based on the corresponding result of Corstanje et al. (2023, Proposition 3.5). We reproduce the proof and clarify the connection between the generator condition and almost sure boundedness of the Lyapunov function.

Proposition 4.2. (*Lyapunov criterion for boundedness*)

Suppose that $V \in D(\mathcal{A}^\circ)$, $V(t, x) \geq 0$ for all (t, x) , $\mathcal{A}^\circ V(t, x) \leq 0$, and $\mathbb{E}^\circ[V(t_0, X_0)] < \infty$. Then, there exists a finite random variable C such that

$$\sup_{t_0 \leq t < T} V(t, X_t) \leq C$$

$\mathbb{P}^\circ$ -almost surely. Furthermore, for every $\lambda > 0$, $\mathbb{P}^\circ(C \geq \lambda) \leq \frac{\mathbb{E}^\circ[V(t_0, X_{t_0})]}{\lambda}$.

Proof:

We will use Doob's maximal inequality for a non-negative supermartingale which says that if M_t is a non-negative supermartingale then for all $\lambda > 0$,

$$\lambda \mathbb{P}^\circ \left[\sup_{t_0 \leq t < T} M_t \geq \lambda \right] \leq \mathbb{E}^\circ[M_{t_0}].$$

In order to apply Doob's maximal inequality, we need to verify its assumptions using Dynkin's formula

$$V(t, X_t) = V(t_0, X_{t_0}) + M_t^v + \int_{t_0}^t \mathcal{A}^\circ V(s, X_s) ds$$

where M_t^v is a local martingale, as defined in Section 2.2.

By assumption, $\mathcal{A}^\circ V \leq 0$ hence $\int_{t_0}^t \mathcal{A}^\circ V(s, X_s) ds \leq 0$. Therefore,

$$V(t, X_t) \leq V(t_0, X_{t_0}) + M_t^v$$

and as $V(t, X_t) \geq 0$ by assumption then

$$M_t^v \geq -V(t_0, X_{t_0}).$$

Since M_t^v is a local martingale that is bounded below by the integrable random variable $-V(t_0, X_{t_0})$, we can conclude that it is a supermartingale by Corstanje et al. (2023, Lemma A.4).

It remains to show that $V(t, X_t)$ is a supermartingale:

1- As $V(t, X_t) \geq 0$, we can write $\mathbb{E}^\circ[V(t, X_t)] = \mathbb{E}^\circ(V(t, X_t))$ and we have just shown that $V(t, X_t) \leq V(t_0, X_{t_0}) + M_t^v$, hence

$$\mathbb{E}^\circ[V(t, X_t)] \leq \mathbb{E}^\circ[V(t_0, X_{t_0})] + \mathbb{E}^\circ[M_t^v]$$

which is integrable as M_t^v is a supermartingale and the initial condition $\mathbb{E}^\circ[V(t_0, X_{t_0})]$ is a constant.

2- In order to show $\mathbb{E}^\circ[V(t, X_t)|\mathcal{F}_s] \leq V(s, X_s)$, we again use Dynkin's formula where we first subtract $V(s, X_s)$ on both sides

$$\begin{aligned} V(t, X_t) - V(s, X_s) &= V(t_0, X_{t_0}) + M_t^v + \int_{t_0}^t \mathcal{A}^\circ V(s, X_s) ds - V(s, X_s) \\ &= V(t_0, X_{t_0}) + M_t^v + \int_{t_0}^t \mathcal{A}^\circ V(s, X_s) ds - V(t_0, X_{t_0}) \\ &\quad - M_s^v - \int_{t_0}^s \mathcal{A}^\circ V(u, X_u) du \\ &\stackrel{s \leq t}{=} M_t^v - M_s^v + \int_s^t \mathcal{A}^\circ V(u, X_u) du. \end{aligned}$$

then by taking expectation on both sides,

$$\mathbb{E}_t^\circ[V(t, X_t)|\mathcal{F}_s] - V(s, X_s) = \mathbb{E}_t^\circ[M_t^v|\mathcal{F}_s] - M_s^v + \mathbb{E}_t^\circ\left[\int_s^t \mathcal{A}^\circ V(u, X_u) du\right] \leq 0$$

where $\mathbb{E}_t^\circ[M_t^v|\mathcal{F}_s] - M_s^v \leq 0$ since M_t^v is a supermartingale and $\mathbb{E}_t^\circ[\int_s^t \mathcal{A}^\circ V(u, X_u) du] \leq 0$ as $\mathcal{A}^\circ V \leq 0$ by assumption.

Therefore, $\mathbb{E}_t^\circ[V(t, X_t)|\mathcal{F}_s] \leq V(s, X_s)$ and thus $V(t, X_t)$ a supermartingale.

To conclude the proof, we apply Doob's maximal inequality:

For all $\lambda > 0$,

$$\mathbb{P}^\circ\left[\sup_{t_0 \leq t < T} V(t, X_t) \geq \lambda\right] \leq \frac{\mathbb{E}_{t_0}^\circ[V(t_0, X_0)]}{\lambda}$$

where $\frac{\mathbb{E}_{t_0}^\circ[V(t_0, X_0)]}{\lambda} \rightarrow 0$ as $\lambda \rightarrow \infty$ and

$$\mathbb{P}^\circ\left(\sup_{t_0 \leq t < T} V(t, X_t) = \infty\right) = \lim_{\lambda \rightarrow \infty} \mathbb{P}^\circ\left[\sup_{t_0 \leq t < T} V(t, X_t) \geq \lambda\right] \leq \lim_{\lambda \rightarrow \infty} \frac{\mathbb{E}_{t_0}^\circ[V(t_0, X_0)]}{\lambda} = 0$$

hence $\mathbb{P}^\circ(\sup_{t_0 \leq t < T} V(t, X_t) = \infty) = 0$.

By Definition 4.1, a random variable exists so that $\sup_{t_0 \leq t < T} V(t, X_t) \leq c$ $\mathbb{P}^\circ$ -almost surely, where $\mathbb{P}^\circ(c \geq \lambda) \leq \lambda^{-1} \mathbb{E}^\circ[V(t_0, X_{t_0})]$. $\blacksquare$

Combining Propositions 4.1 and 4.2 gives $\mathcal{A}^\circ V \leq 0$ which implies that $V(t, X_t)$ is a non-negative supermartingale, which in turn implies that $\sup_{t_0 \leq t < T} V(t, X_t)$ is finite $\mathbb{P}^\circ$ -almost surely and hence $\mathbb{P}_T^\circ(A(T)) = 1$.

If V is also chosen so that boundedness of $V(t, X_t)$ on $A_k(t)$ provides a bound on $\Psi_t(X)\kappa(t)$, then the local boundedness required in Assumption 4.5

is satisfied. The Lyapunov function provides a construction of the localising events for which $A(T)$ has full probability under the guided law, and it can be used to control the correction factor on these events. In Chapter 5, we construct a specific Lyapunov function for the one-dimensional diffusion case and verify these properties in that setting.

4.3 Terminal Correction Factor and Terminal Time Theorem

For $t < T$, the Radon-Nikodym derivative in Proposition 3.1 contains the correction factor $\Psi_t(X)$. To extend this to the terminal time, we require a corresponding terminal quantity. Since the behaviour near time T is controlled only on the localising events introduced in Section 4.1, we consider the quantities

$$\Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)}.$$

The following lemma, corresponding to the terminal limit result of Corstanje et al. (2023, Lemma 3.2), shows that these quantities converge to a well defined limit as $t \uparrow T$, giving the terminal quantity $\Psi_T^\kappa(X)$. With the additional terminal limit condition in Assumption 4.5, we show how the localising events $A_k(t)$ are used to obtain the terminal correction factor.

Lemma 4.1. *Define $A_k(T) = \bigcap_{t < T} A_k(t)$.*

Under Assumptions 4.1-4.5, there exists a random variable $\Psi_T^\kappa(X)$ so that for all k ,

$$\Psi_T^\kappa(X)\mathbf{1}_{A_k(T)} = \lim_{t \uparrow T} \Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)}.$$

Proof:

By Assumption 4.1, $A_k(t)$ decreases as t increases, and $A_k(T) = \bigcap_{t < T} A_k(t)$ hence $A_k(T) \subseteq A_k(t)$ for all t .

By Assumption 4.5, for each fixed k , $\Psi_t(X)\kappa(t)$ is uniformly bounded on $A_k(t)$, and hence also on $A_k(T) \subseteq A_k(t)$. By the terminal limit condition in Assumption 4.5, there exists a random variable $\Psi_t^\kappa(X)$ such that

$$\lim_{t \uparrow T} \Psi_t(X)\kappa(t)\mathbf{1}_{A_k(T)} = \Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}$$

$\mathbb{P}^\circ$ -almost surely. We now can show $\lim_{t \uparrow T} \Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)} = \Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}$.

As $A_k(t) = A_k(T) \cup A_k(t) \setminus A_k(T)$,

$$\begin{aligned} |\Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)} - \Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}| &= |\Psi_t(X)\kappa(t)\mathbf{1}_{A_k(T) \cup A_k(t) \setminus A_k(T)} - \Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}| \\ &\leq |\Psi_t(X)\kappa(t) - \Psi_T^\kappa(X)|\mathbf{1}_{A_k(T)} \\ &\quad + |\Psi_t(X)\kappa(t)|\mathbf{1}_{A_k(t) \setminus A_k(T)}. \end{aligned}$$

The first term on the RHS converges to zero as $\lim_{t \uparrow T} \Psi_t(X)\kappa(t)\mathbf{1}_{A_k(T)} = \Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}$ and the second term on the RHS converges to zero as well since it is bounded by Assumption 4.5 and the events shrink. Hence, the LHS converges to zero, concluding the proof. $\blacksquare$

Theorem 4.1 follows the proof strategy of Corstanje et al. (2023, Theorem 3.3). Several steps in their proof are stated only briefly, so here the reasoning is reformulated, reorganised to clarify the limiting arguments, the role of the localising events, and the use of the preceding assumptions.

Theorem 4.1. *Suppose Assumptions 4.1-4.5 are satisfied, and let $\Psi_T^\kappa(X)$ be the terminal correction factor defined in Lemma 4.1. Then, for every bounded, measurable f ,*

$$\mathbb{E}_T^*(f(X)\mathbf{1}_{A(T)}) = \mathbb{E}_T^\circ(f(X)\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A(T)})$$

where

$$\bar{h} = \frac{\tilde{h}(0, x_0)}{h(0, x_0)}.$$

In particular, if $\mathbb{P}_T^\circ(A(T)) = 1$, then $\mathbb{P}_T^* \ll \mathbb{P}_T^\circ$ and

$$\frac{d\mathbb{P}_T^*}{d\mathbb{P}_T^\circ}(X) = \bar{h}\Psi_T^\kappa(X).$$

Note that Corstanje et al. (2023, Theorem 3.3) state the corresponding result as an equivalence of the conditioned and guided measures. Here, we first show the terminal change of measure result and the Radon-Nikodym derivative together with $P_T^* \ll P_T^\circ$. The reverse absolute continuity is discussed separately after the proof, as it is not made explicit that $\bar{h}\Psi_T^\kappa(X) > 0$.

The proof proceeds by first identifying the expectation of the terminal correction factor on $A(T)$. We then show that the corresponding finite time quantities converge to this value in order to satisfy the assumptions of Scheffé's lemma to apply it. The lemma gives $L^1(\mathbb{P}^\circ)$ convergence of the likelihood ratios, which allows the finite time change of measure formula from Proposition 3.1 to be passed to the terminal time.

The following lemma follows directly from Scheffe's theorem for non-negative random variables as stated in Billingsley (1995).

Lemma 4.2. (*Scheffe's lemma*)

Consider Y_t to be a random variable indexed by time. If $Y_t \geq 0$, Y_t converges to Y in the almost sure sense, and $\mathbb{E}(Y_t)$ converges to $\mathbb{E}(Y)$ then Y_t converges to Y in L^1 , as $t \uparrow T$.

Proof of Theorem 4.1:

1- Establishing the limiting expectation on $A(T)$ i.e.

$$\mathbb{E}_T^\circ(\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A(T)}) = \mathbb{P}_T^*(A(T)).$$

For fixed k , Lemma 4.1 gives

$$\lim_{t \uparrow T} \bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)} = \bar{h}\Psi_T^\kappa\mathbf{1}_{A_k(T)}.$$

Together with Assumption 4.5, dominated convergence gives

$$\mathbb{E}_T^\circ[\bar{h}\Psi_T^\kappa\mathbf{1}_{A_k(T)}] = \lim_{t \uparrow T} \mathbb{E}_t^\circ[\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)}]. \quad (4.1)$$

For $t < T$, Proposition 3.1 gives

$$\frac{d\mathbb{P}_t^*}{d\mathbb{P}_t^\circ} = \bar{h}\Psi_t \frac{h}{\bar{h}}$$

and using the change of measure formula, as defined in Section 2.3, for every bounded measurable f ,

$$\mathbb{E}_t^*(f(X)) = \mathbb{E}_t^\circ \left(f(X) \frac{\tilde{h}(0, x_0)}{h(0, x_0)} \Psi_t(X) \frac{h(t, X_t)}{\tilde{h}(t, X_t)} \right). \quad (4.2)$$

Letting $f(X) = \frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A_k(t)}$, and plugging it into (4.2) yields

$$\begin{aligned} \mathbb{E}_t^* \left(\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A_k(t)} \right) &= \mathbb{E}_t^\circ \left(\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A_k(t)} \frac{\tilde{h}(0, x_0)}{h(0, x_0)} \Psi_t(X) \frac{h(t, X_t)}{\tilde{h}(t, X_t)} \right) \\ &= \mathbb{E}_t^\circ \left(\kappa(t) \mathbf{1}_{A_k(t)} \frac{\tilde{h}(0, x_0)}{h(0, x_0)} \Psi_t(X) \right) \\ &= \mathbb{E}_t^\circ (\kappa(t) \mathbf{1}_{A_k(t)} \bar{h}\Psi_t(X)) \end{aligned} \quad (4.3)$$

Plugging (4.3) in (4.1):

$$\mathbb{E}_T^\circ[\bar{h}\Psi_T^\kappa\mathbf{1}_{A_k(T)}] = \lim_{t \uparrow T} \mathbb{E}_t^\circ[\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)}] = \lim_{t \uparrow T} \mathbb{E}_t^* \left(\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A_k(t)} \right).$$

Taking the limit of $\mathbb{E}_T^\circ(\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A_k(T)})$ as $k \rightarrow \infty$ together with Assumption 4.1, monotone convergence gives

$$\lim_{k \rightarrow \infty} \mathbb{E}_T^\circ(\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}) = \mathbb{E}_T^\circ(\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{\bigcup_{k=1}^\infty A_k(T)}).$$

By the proof of Proposition 4.1, we established that $\bigcup_{k=1}^\infty A_k(T) = A(T)$, P_T° -almost surely. Therefore,

$$\lim_{k \rightarrow \infty} \mathbb{E}_T^\circ(\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}) = \mathbb{E}_T^\circ(\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A(T)}).$$

Hence, we obtain

$$\begin{aligned} \mathbb{P}_T^*(A(T)) &\stackrel{\text{Assumption (4.4)}}{=} \lim_{k \rightarrow \infty} \lim_{t \uparrow T} \mathbb{E}_t^* \left(\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A_k(t)} \right) \\ &= \lim_{k \rightarrow \infty} \mathbb{E}_T^\circ(\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}) \\ &= \mathbb{E}_T^\circ(\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A(T)}). \end{aligned}$$

2- Show that the finite time expectations converge to the same value i.e. $\lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) = \mathbb{P}_T^*(A(T))$.

We start by showing

$$\lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) \geq \mathbb{P}_T^*(A(T)). \quad (4.4)$$

From Assumptions (4.1) and (4.4), $A_k(t) \subseteq A_{k+1}(t)$ and taking $k \rightarrow \infty$, $A(t) = \bigcup_{k \geq 1} A_k(t)$. Thus, for any fixed m ,

$$A_m(t) \subseteq \bigcup_{k \geq 1} A_k(t) = A(t)$$

Therefore, $\mathbf{1}_{A_m(t)} \leq \mathbf{1}_{A(t)}$. Hence,

$$\begin{aligned} \lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) &\geq \lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A_m(t)}) \\ &\stackrel{(4.3)}{=} \lim_{t \uparrow T} \mathbb{E}_t^* \left(\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A_m(t)} \right). \end{aligned}$$

As $m \rightarrow \infty$, and using Assumption 4.4:

$$\lim_{m \rightarrow \infty} \lim_{t \uparrow T} \mathbb{E}_t^* \left(\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A_k(t)} \right) = \mathbb{P}_T^*(A(T))$$

gives us

$$\lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) \geq \lim_{m \rightarrow \infty} \lim_{t \uparrow T} \mathbb{E}_t^* \left(\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A_m(t)} \right) = \mathbb{P}_T^*(A(T)).$$

For the other inequality,

$$\lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) \leq \mathbb{P}_T^*(A(T)). \quad (4.5)$$

Assumption 4.1 says for $s \leq t$, $A(t) \subseteq A(s)$ and thus, $\mathbf{1}_{A(t)} \leq \mathbf{1}_{A(s)}$. Hence,

$$\begin{aligned} \lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) &\leq \lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(s)}) \\ &\stackrel{(4.3)}{=} \lim_{t \uparrow T} \mathbb{E}_t^* \left(\frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A(s)} \right) \\ &= \mathbb{P}_s^*(A(s)) \end{aligned}$$

where the last equality follows from the following lemma by considering $g_s(X) = \mathbf{1}_{A(s)}$, adapted from Corstanje et al. (2023, Lemma A.1).

Lemma 4.3. *Suppose Assumptions (4.2) and (4.3) hold. Let $s < T$ and let g_s be a bounded, $\mathcal{F}_s$ measurable function. Then,*

$$\lim_{t \uparrow T} \mathbb{E}_t^* \left(g_s(X) \kappa(t) \frac{\tilde{h}(t, X_t)}{h(t, X_t)} \right) = \mathbb{E}_s^*(g_s(X)).$$

The inequality $\lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) \leq \mathbb{P}_s^*(A(s))$ holds for every s , therefore

$$\lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) \leq \inf_{s < T} \mathbb{P}_s^*(A(s)) = \mathbb{P}_T^*(A(T)) \quad (4.6)$$

proving (4.5).

Note that the equality in (4.6) comes from taking the limit $s \rightarrow T$, and by Assumption 4.1 the events decrease to the limiting event i.e. $A(s) \rightarrow A(T)$, where by Assumption 4.4 the limit is $A(T) = \bigcap_{s < T} A(s)$. Using continuity of measures, we obtain

$$\lim_{s \rightarrow T} \mathbb{P}_s^*(A(s)) = \mathbb{P}_T^*(A(T)).$$

Combining both (4.4) and (4.5), we have

$$\lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) = \mathbb{P}_T^*(A(T)).$$

3- Apply Scheffe's lemma (Lemma 4.2).

Combining the first two steps,

$$\lim_{t \uparrow T} \mathbb{E}_t^\circ(\bar{h}\Psi_t\kappa(t)\mathbf{1}_{A(t)}) = \mathbb{P}_T^*(A(T)) = \mathbb{E}_T^\circ(\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A(T)}) \quad (4.7)$$

Corstanje et al. (2023) obtain the corresponding convergence by passing from the fixed k result to $A(t)$ through an interchange of the limits $k \rightarrow \infty$ and $t \uparrow T$. Since monotonicity in k alone does not in general justify an interchange of these two limits, we avoid this step here. We instead use the proof of Proposition 4.1, where we have seen that for $\mathbb{P}^\circ$ -almost every path ω , there exists a finite integer $k = k(\omega)$ such that $\omega \in A_k(t)$ for every $t < T$. Hence, $\omega \in A_k(T)$. For this fixed k , Lemma 4.1 gives

$$\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A_k(t)} \rightarrow \bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}.$$

On such a path, as $\omega \in A_k(t)$, and $A_k(t) \subseteq A(t)$ then $\omega \in A(t)$ and $\mathbf{1}_{A_k(t)} = \mathbf{1}_{A(t)} = 1$ for every $t < T$, while $\mathbf{1}_{A_k(T)} = \mathbf{1}_{A(T)} = 1$ as $\omega \in A_k(T)$, and $A_k(T) \subseteq A(T)$ thus $\omega \in A(T)$. Therefore,

$$\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A(t)} \rightarrow \bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A(T)}$$

$\mathbb{P}^\circ$ -almost surely, as $t \uparrow T$.

As we have shown (4.7) and $\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A(t)} \rightarrow \bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A_k(T)}$ as $t \uparrow T$, we can apply Scheffe's lemma to obtain

$$\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A(t)} \rightarrow \bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A(T)}$$

in $L^1(\mathbb{P}^\circ)$ as $t \uparrow T$.

4- Pass the finite time change of measure identity to T i.e. $\mathbb{E}_T^*[f(X)\mathbf{1}_{A(T)}] = \mathbb{E}_T^\circ[f(X)\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A(T)}]$ for every bounded measurable f .

Let $s < T$ and f_s be any bounded, positive, $\mathcal{F}_s$ measurable function such that the support of $f_s(X)$ is contained in $A(T)$ i.e. $f_s(X(\omega)) = 0$ when $\omega \notin A(T)$.

Since we have shown that $\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A(t)} \rightarrow \bar{h}\Psi_T^\kappa(X)$ in L^1 and f_s is bounded by assumption, then

$$f_s(X)\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A(t)} \rightarrow f_s(X)\bar{h}\Psi_T^\kappa(X)$$

in L^1 too, thus

$$\mathbb{E}_t^\circ(f_s(X)\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A(t)}) \rightarrow \mathbb{E}_T^\circ(f_s(X)\bar{h}\Psi_T^\kappa(X))$$

hence,

$$\begin{aligned}\mathbb{E}_T^\circ(f_s(X)\bar{h}\Psi_T^\kappa(X)) &= \lim_{t \uparrow T} \mathbb{E}_t^\circ(f_s(X)\bar{h}\Psi_t(X)\kappa(t)\mathbf{1}_{A(t)}) \\ &\stackrel{(4.3)}{=} \lim_{t \uparrow T} \mathbb{E}_t^* \left(f_s(X) \frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A(t)} \right).\end{aligned}\quad (4.8)$$

As $f_s(X)$ has support only on $A(T)$ and by Assumption 4.1, $t \in [s, T)$, $A(T) \subseteq A(t) \subseteq A(s)$, hence $\mathbf{1}_{A(T)} \leq \mathbf{1}_{A(t)} \leq \mathbf{1}_{A(s)}$, then $f_s(X)\mathbf{1}_{A(T)} = f_s(X)\mathbf{1}_{A(t)} = f_s(X)\mathbf{1}_{A(s)}$.

Using this in (4.9),

$$\begin{aligned}\mathbb{E}_T^\circ(f_s(X)\bar{h}\Psi_T^\kappa(X)) &= \lim_{t \uparrow T} \mathbb{E}_t^* \left(f_s(X) \frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A(t)} \right) \\ &= \lim_{t \uparrow T} \mathbb{E}_t^* \left(f_s(X) \frac{\tilde{h}(t, X_t)}{h(t, X_t)} \kappa(t) \mathbf{1}_{A(s)} \right) \\ &\stackrel{\text{Lemma 4.3}}{=} \mathbb{E}_s^*(f_s(X)\mathbf{1}_{A(s)}) \\ &= \mathbb{E}_T^*(f_s(X)\mathbf{1}_{A(T)})\end{aligned}$$

where the last equality follows from the consistency of family of measures.

5- Finally, we show that $A(T)$ has probability one under the conditioned measure. Since $P_T^\circ(A(T)) = 1$ is assumed, showing $P_T^*(A(T)) = 1$ allows the indicator $\mathbf{1}_{A(T)}$ to be removed from both sides of the localised change of measure identity. This gives us the Radon-Nikodym derivative on the full path space and therefore establishes $P_T^* \ll P_T^\circ$.

By Assumptions 4.1 and 4.4 the family $(A(t))_{t < T}$ is decreasing. Let $(t_n)_{n \in \mathbb{N}}$ be a sequence such that $t_n \uparrow T$, then $A(t_n) \downarrow A(T)$ where $A(T) = \bigcap_{n=1}^\infty A(t_n)$. By continuity from above

$$\mathbb{P}_T^*(A(T)) = \lim_{n \rightarrow \infty} \mathbb{P}_T^*(A(t_n)).$$

For every n , $A(T) \subseteq A(t_n)$, and monotonicity of probability gives

$$\mathbb{P}_T^\circ(A(t_n)) \geq \mathbb{P}_T^\circ(A(T)) = 1$$

where $\mathbb{P}_T^\circ(A(T)) = 1$ follows by assumption. Therefore, $\mathbb{P}_T^\circ(A(t_n)) = 1$ for every $t < T$. Since $A(t_n) \in \mathcal{F}_{t_n}$ and $\mathbb{P}_{t_n}^\circ$ is the restriction of $\mathbb{P}_T^\circ$ to $\mathcal{F}_{t_n}$,

$$\mathbb{P}_{t_n}^\circ(A(t_n)) = \mathbb{P}_T^\circ(A(t_n)) = 1.$$

By Proposition 3.1,

$$\mathbb{P}_{t_n}^* \sim \mathbb{P}_{t_n}^\circ$$

meaning for any event B ,

$$\mathbb{P}_{t_n}^*(B) = 0 \iff \mathbb{P}_{t_n}^\circ(B) = 0.$$

Taking $B = A(t_n)^c$, $\mathbb{P}_{t_n}^\circ(A(t_n)^c) = 0$ as $\mathbb{P}_{t_n}^\circ(A(t_n)) = 1$ and by equivalence, $\mathbb{P}_{t_n}^*(A(t_n)^c) = 0$ therefore $\mathbb{P}_{t_n}^*(A(t_n)) = 1$, which holds for every n . Hence,

$$\mathbb{P}_T^*(A(T)) = \lim_{n \rightarrow \infty} \mathbb{P}_T^*(A(t_n)) = \lim_{n \rightarrow \infty} 1 = 1.$$

concluding the proof of the theorem. $\square$

Corstanje et al. (2023) state equivalence, but the final argument displayed in the proof establishes that the localisation event $A(T)$ has full probability under both measures. On its own, this does not establish the reverse absolute continuity $P_T^\circ \ll P_T^*$. As both measures give $A(T)$ probability one, then by the statement of the proof

$$\mathbb{E}_T^*[f(X)\mathbf{1}_{A(T)}] = \mathbb{E}_T^\circ[f(X)\bar{h}\Psi_T^\kappa(X)\mathbf{1}_{A(T)}].$$

Taking $B \in \mathcal{F}_T$ and choosing $f = \mathbf{1}_B$. Then,

$$\begin{aligned} \mathbb{P}_T^*(B \cap A(T)) &= \int_{B \cap A(T)} \bar{h}\Psi_T^\kappa d\mathbb{P}_T^\circ \\ \implies \mathbb{P}_T^*(B) &= \int_B \bar{h}\Psi_T^\kappa d\mathbb{P}_T^\circ \end{aligned}$$

for every $B \in \mathcal{F}_T$.

Let $Z_T = \bar{h}\Psi_T^\kappa$, $\mathbb{P}_T^\circ$ -almost surely. In order to show the reverse absolute continuity we consider $\mathbb{P}_T^*(B) = 0$, hence

$$\mathbb{P}_T^*(B) = \int_B Z_T d\mathbb{P}_T^\circ = 0$$

and in order to get $\mathbb{P}_T^\circ \ll \mathbb{P}_T^*$, we would require $\mathbb{P}_T^\circ(B) = 0$, and $Z_T > 0$.

However, showing that Z_T is strictly positive in the general Markov process setting can be technically difficult. If we consider instead the diffusion setting, the structure of the process makes it easier to verify this positivity condition. Whereas we immediately obtain $\mathbb{P}_T^* \ll \mathbb{P}_T^\circ$ from the proof of Theorem 4.1 since $\mathbb{P}_T^\circ(B) = 0$ implies $\mathbb{P}_T^*(B) = 0$ regardless of the value of Z_T on B .

Chapter 5

Application to a One-Dimensional Conditioned Diffusion

This chapter applies the general Markov Process framework in Chapter 4 to a one-dimensional conditioned diffusion. We state the assumptions required for terminal time equivalence and examine how localisation is implemented through a Lyapunov function. We look at the choice of the logarithmic Lyapunov function used to control the singular behaviour as $t \uparrow T$. Finally, we connect the classical guided proposal framework with neural guided diffusion bridges and illustrate the resulting variational correction in a simple Brownian bridge example.

5.1 A One-Dimensional Bridge Model

The broader guided proposal framework allows for multidimensional, possibly hypoelliptic diffusions and partial terminal observations of the form $LX_T = v$ (Bierkens et al., 2020; Corstanje et al., 2023). In this chapter, we develop a one-dimensional, uniformly elliptic diffusion and fully observed specialisation with terminal condition $X_T = v$ and a linear auxiliary process with constant coefficients, we also consider $\kappa(t) = 1$ as mentioned in Section 4.1.

As discussed in Chapter 3, consider the true diffusion

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t$$

with initial condition $X_0 = x_0$, and the conditioned process has the SDE $X^* = (X \mid X_T = v)$ with drift

$$b^*(t, x) = b(t, x) + a(t, x) \nabla_x \log h(t, x)$$

where $h(t, x) = p(t, x; T, v)$ is determined from the transition density of the true diffusion and is generally intractable. Consider the auxiliary process

$$d\tilde{X}_t = (\tilde{B}\tilde{X}_t + \tilde{\beta})dt + \tilde{\sigma}dW_t$$

where $\tilde{B}, \tilde{\beta} \in \mathbb{R}$. As everything is constant, this is the one-dimensional Ornstein-Uhlenbeck process, or when $\tilde{B} = 0$, it is Brownian motion with a drift.

Unlike the general formulation, where the corresponding quantities are introduced directly, in the one-dimensional setting we derive them from the auxiliary SDE. This provides a clearer interpretation of their probabilistic meaning and terminal behaviour.

$$d\tilde{X}_s = (\tilde{B}\tilde{X}_s + \tilde{\beta})ds + \tilde{\sigma}dW_s, \quad \tilde{B} \neq 0.$$

Multiplying both sides by the integrating factor $e^{-\tilde{B}t}$ and integrating from t to T

$$\begin{aligned} d(e^{-\tilde{B}s}\tilde{X}_s) &= -\tilde{B}e^{-\tilde{B}s}\tilde{X}_s ds + e^{-\tilde{B}s}d\tilde{X}_s \\ &= -\tilde{B}e^{-\tilde{B}s}\tilde{X}_s ds + e^{-\tilde{B}s}[(\tilde{B}\tilde{X}_s + \tilde{\beta})ds + \tilde{\sigma}dW_s] \\ &= e^{-\tilde{B}s}\tilde{\beta}ds + e^{-\tilde{B}s}\tilde{\sigma}dW_s \\ \implies e^{-\tilde{B}T}\tilde{X}_T - e^{-\tilde{B}t}\tilde{X}_t &= \tilde{\beta} \int_t^T e^{-\tilde{B}s}ds + \tilde{\sigma} \int_t^T e^{-\tilde{B}s}dW_s. \end{aligned}$$

Multiplying by $e^{\tilde{B}T}$ and conditioning on $\tilde{X}_t = x$

$$\tilde{X}_T = e^{\tilde{B}(T-t)}x + \int_t^T \tilde{\beta}e^{\tilde{B}(T-s)}ds + \int_t^T e^{\tilde{B}(T-s)}\tilde{\sigma}dW_s.$$

We now define

$$l(t) := e^{\tilde{B}(T-t)} \tag{5.1}$$

and

$$\mu(t) := \tilde{\beta} \int_t^T e^{\tilde{B}(T-s)}ds \tag{5.2}$$

where $l(t)$ describes how the linear drift $\tilde{B}\tilde{X}_s$ changes the current state x over the time interval from t to T and $\mu(t)$ represents the total contribution of the additional drift $\tilde{\beta}$ accumulated between times t and T .

The predicted terminal value of the auxiliary process is

$$\mathbb{E}[\tilde{X}_T \mid \tilde{X}_t = x] = l(t)x + \mu(t)$$

as the expectation of the stochastic integral is zero. The endpoint error residual would be

$$\zeta(t, x) = v - \mathbb{E}[\tilde{X}_T \mid \tilde{X}_t = x] = v - l(t)x - \mu(t). \quad (5.3)$$

The variance is

$$\text{Var}(\tilde{X}_T \mid \tilde{X}_t = x) = \tilde{a} \int_t^T e^{2\tilde{B}(T-s)} ds$$

where $\tilde{a} = \tilde{\sigma}^2$ and after solving the integral, we obtain

$$\text{Var}(\tilde{X}_T \mid \tilde{X}_t = x) = \tilde{a} \frac{e^{2\tilde{B}(T-t)} - 1}{2\tilde{B}}.$$

We define the precision

$$M(t) := \text{Var}(\tilde{X}_T \mid \tilde{X}_t = x)^{-1}.$$

Since the precision is the reciprocal of the variance, a larger value of $M(t)$ corresponds to a smaller uncertainty in the predicted terminal position. From the conditional variance derived above,

$$M(t) = \frac{2\tilde{B}}{\tilde{a}(e^{2\tilde{B}(T-t)} - 1)} \quad (5.4)$$

We further define

$$H(t, x) := \frac{1}{2} M(t) \zeta(t, x)^2 \quad (5.5)$$

Recall that $\zeta(t, x)$ from (5.3) is the difference between the desired terminal endpoint v and the conditional mean of $\tilde{X}_T$ given $\tilde{X}_t = x$. Hence, $H(t, x)$ measures the squared terminal residual relative to the conditional uncertainty of the auxiliary process. As $t \uparrow T$, the conditional variance decreases and $M(t)$ increases, so a fixed terminal residual is penalised strongly.

Since $\tilde{X}_T \mid \tilde{X}_t \sim N(l(t)x + \mu(t), M(t)^{-1})$, the auxiliary conditioning function $\tilde{h}(t, x) = \tilde{p}(t, x; T, v)$ is given by

$$\tilde{h}(t, x) = \sqrt{\frac{M(t)}{2\pi}} \exp\{-H(t, x)\} \quad (5.6)$$

Thus, a small value of $H(t, x)$ means that the desired endpoint v is close to the terminal value predicted by the auxiliary process relative to its remaining uncertainty, whereas a large value of $H(t, x)$ indicates a large terminal discrepancy.

We can now compute $\tilde{r}(t, x)$ and obtain the guided process

$$\begin{aligned}\tilde{r}(t, x) &= \partial_x \log \tilde{h}(t, x) = -\partial_x H(t, x) \\ &= -\partial_x \left(\frac{1}{2} M(t) \zeta(t, x)^2 \right) \\ &= -\partial_x \left(\frac{1}{2} M(t) (v - \mu(t) - l(t)x)^2 \right) \\ &= l(t) M(t) (v - \mu(t) - l(t)x) = l(t) M(t) \zeta(t, x).\end{aligned}$$

Consequently, the guided process is

$$dX_t^\circ = [b(t, X_t^\circ) + a(t, X_t^\circ) l(t) M(t) (v - \mu(t) - l(t) X_t^\circ)] dt + \sigma(t, X_t^\circ) dW_t^\circ.$$

Hence, the guidance term that guides our process towards the terminal endpoint v is $a(t, X_t^\circ) l(t) M(t) (v - \mu(t) - l(t) X_t^\circ)$. Recall that X° retains the drift and diffusion coefficients from the true process, only the guiding term is obtained from the auxiliary process.

It can also be seen that near T , $l(t) \rightarrow 1$, $\mu(t) \rightarrow 0$, $M(t) \sim \frac{1}{\tilde{a}(T-t)}$. Hence

$$a(t, x) \tilde{r}(t, x) \sim \frac{a(t, x)}{\tilde{a}} \frac{v - x}{T - t}.$$

Thus, the guiding drift has the characteristic singular behaviour of a Brownian bridge near the terminal time. In particular, when the diffusion coefficients of the true and auxiliary processes agree, $a(t, x) = \tilde{a}$, this reduces to $\frac{v-x}{T-t}$ which is the standard Brownian bridge pull towards v .

5.2 Assumptions and Main Theorem

The following assumptions are obtained by specialising the diffusion conditions of Corstanje et al. (2023) to the one-dimensional model introduced above. Their original conditions are stated in a more general setting. Here, we derive the corresponding scalar forms and give an interpretation of each condition in terms of the quantities $l(t)$, $M(t)$, $\zeta(t, x)$, and the coefficients of the true and auxiliary diffusion.

The assumptions justify extending the finite time change of measure result, valid for $t < T$, to the terminal time T . As noted in Section 5.1,

$$M(t) \sim \frac{1}{\tilde{a}(T-t)}$$

as $t \uparrow T$ so the guiding drift becomes singular near the terminal time. The assumptions below ensure that this singular behaviour remains sufficiently controlled for the likelihood ratio to admit a finite limit and for the guided process to reach the endpoint.

Let $t_0 < T$, we assume that the coefficients of X and $\tilde{X}$ satisfy the following conditions for $t \in [t_0, T]$.

Assumption 5.1. (*Correct terminal order of the auxiliary precision*)

There exist constants $0 < c \leq \bar{c} < \infty$ such that

$$\frac{c}{T-t} \leq M(t) \leq \frac{\bar{c}}{T-t}$$

for t sufficiently close to T .

This assumption says that the auxiliary process should shrink uncertainty at the same rate as a normal diffusion bridge. In the linear auxiliary model, this condition can be verified directly using Taylor expansion. As

$$M(t)^{-1} = \tilde{a} \int_t^T e^{2\tilde{B}(T-s)} ds.$$

Performing change of variable, taking $u = T - s$ then $du = -ds$,

$$= \tilde{a} \int_0^{T-t} -e^{2\tilde{B}u} du = \tilde{a} \int_0^{T-t} e^{2\tilde{B}u} du.$$

By Taylor expanding $e^{2\tilde{B}u}$ as $u \rightarrow 0$, $e^{2\tilde{B}u} = 1 + 2\tilde{B}u + O(u^2)$. Therefore,

$$M(t)^{-1} = \tilde{a} \int_0^{T-t} (1 + 2\tilde{B}u + O(u^2)) du = \tilde{a}(T-t) + \tilde{a}\tilde{B}(T-t)^2 + O((T-t)^3).$$

In particular, $M(t)^{-1} = \tilde{a}(T-t)[1 + O(T-t)]$ thus,

$$M(t) = \frac{1}{\tilde{a}(T-t)}[1 + O(T-t)] \sim \frac{1}{\tilde{a}(T-t)}$$

as $t \uparrow T$. Equivalently,

$$(T-t)M(t) \rightarrow \frac{1}{\tilde{a}}$$

as $t \uparrow T$. As $\frac{1}{a} > 0$, it follows that for t sufficiently close to T , there exist constants $0 < c \leq \bar{c} < \infty$ such that $\frac{c}{T-t} \leq M(t) \leq \frac{\bar{c}}{T-t}$ verifying Assumption (5.1).

Assumption 5.2. (*Controlled difference between the drifts*)

There exists a constant $c_1 > 0$ such that

$$|l(t)[\tilde{b}(t, X_t^\circ) - b(t, X_t^\circ)]| \leq c_1, \quad \tilde{b}(t, x) = \tilde{B}x + \tilde{\beta}.$$

with $l(t)$, as defined in (5.1), is bounded above and away from zero on $[t_0, T]$, this condition is equivalent, up to a change in the constant, to requiring

$$|\tilde{b}(t, X_t^\circ) - b(t, X_t^\circ)| \leq c_1.$$

Thus, the difference between the auxiliary and true drift must remain uniformly bounded along the guided trajectory. The two drifts do not need to be equal, however their difference must remain controlled. This assumption is used to control the drift dependent terms that appear in the correction factor and prevents them from creating additional integrability problems as $t \uparrow T$.

As an example, suppose that the true and auxiliary drifts are affine

$$b(t, x) = Bx + \beta, \quad \tilde{b}(t, x) = \tilde{B}x + \tilde{\beta}.$$

Then,

$$\tilde{b}(t, x) - b(t, x) = (\tilde{B} - B)x + (\tilde{\beta} - \beta).$$

If $\tilde{B} = B$, the discrepancy reduces to the constant $\tilde{\beta} - \beta$, making the boundedness condition immediate.

If $\tilde{B} \neq B$, the discrepancy grows linearly with x , we would need additional control of the guided trajectory to ensure that Assumption 5.2 holds.

Assumption 5.3. (*The projected true diffusivity is uniformly bounded*)

There exists a constant $c_2 > 0$ such that

$$l(t)^2 a(t, X_t^\circ) \leq c_2.$$

Since $l(t)$ is bounded on $[t_0, T]$, Assumption 5.3 is satisfied whenever the true diffusivity $a(t, x)$ is uniformly bounded above. This is separate from uniform ellipticity, which provides a lower bound on $a(t, x)$. Together, the two conditions ensure that the diffusivity remains bounded above and away from zero.

Assumption 5.4. (*Compatibility of the true and auxiliary diffusions near T*)

There exist constants $c_3, c_4, \alpha > 0$ and a non-negative function $\varepsilon : [t_0, T) \times \mathbb{R} \rightarrow [0, \infty)$ such that $\forall t \in [t_0, T)$,

$$l(t)^2 \mid \tilde{a} - a(t, X_t^\circ) \mid \leq c_3 \mid \zeta(t, X_t^\circ) \mid + \varepsilon(t, X_t^\circ)$$

where $\varepsilon(t, x) \leq c_4(T - t)^\alpha$. Hence,

$$l(t)^2 \mid \tilde{a} - a(t, X_t^\circ) \mid \leq c_3 \mid \zeta(t, X_t^\circ) \mid + c_4(T - t)^\alpha.$$

This condition controls the difference between the true and auxiliary diffusions near the terminal time. The discrepancy is bounded by two terms: the terminal residual $\mid \zeta(t, X_t^\circ) \mid$, which measures how far the guided trajectory is from the required endpoint, and a remainder term that vanishes at rate $(T - t)^\alpha$. This is needed because the discrepancy is multiplied by quantities that become singular as $t \uparrow T$, so some decrease is required to keep the terms integrable.

Assumption 5.5. (*Gaussian upper bound on the transition density*)

For $K > 1$ and $u \geq 0$, define

$$g_K(u) = \max \left(\frac{1}{K}, 1 - Ku \right).$$

There exists constants $\Lambda, C > 0$, $K > 1$ and a function $m_t(s, x) : \{s, t : 0 \leq s \leq t \leq T\} \times \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\mid m_t(s, x) - x \mid < K(t - s)\mid x \mid$$

and

$$\mid m_t(s, x) \mid^2 \geq g_K(t - s)\mid x \mid^2$$

so that $\forall s < t \leq T$ and $x, y \in \mathbb{R}$,

$$p(s, y; t, x) \leq C(t - s)^{-1/2} \exp \left[-\frac{\Lambda \mid x - m_t(s, y) \mid^2}{t - s} \right].$$

This assumption requires the transition density of the original diffusion to satisfy a Gaussian upper bound. The factor $(t - s)^{-1/2}$ is the one-dimensional scaling of a diffusion transition density, while the exponential term controls the probability of large deviations from the predicted location $m_t(s, y)$. Such a bound provides control of the transition density near the terminal time and

is used to establish the limiting estimates required for the change of measure result.

Combining the one-dimensional quantities and assumptions derived above gives the following specialisation of Corstanje et al. (2023, Theorem 5.6).

Theorem 5.1. (*One-dimensional specialisation*)

Let X solve the SDE $dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t$, with initial condition $X_0 = x_0$, $X^* = (X \mid X_T = v)$ denote the true conditioned diffusion, and the auxiliary process $d\tilde{X}_t = (\tilde{B}\tilde{X}_t + \tilde{\beta}dt) + \tilde{\sigma}dW_t$ and $l(t), \mu(t), \zeta(t, x), M(t)$, and $\tilde{h}(t, x)$ as defined in (5.1), (5.2), (5.3), (5.4), and (5.6) respectively.

Let the guided process X° solve:

$$dX_t^\circ = [b(t, X_t^\circ) + a(t, X_t^\circ)l(t)M(t)\zeta(t, X_t^\circ)]dt + \sigma(t, X_t^\circ)dW_t^\circ.$$

Suppose that the assumptions above hold. Then, the laws of X^* and X° are equivalent on $[0, T]$ i.e.

$$\mathbb{P}_T^* \sim \mathbb{P}_T^\circ.$$

Furthermore,

$$\frac{d\mathbb{P}_T^*}{d\mathbb{P}_T^\circ}(X) = \frac{\tilde{h}(0, x_0)}{h(0, x_0)} \Psi_T(X)$$

is the likelihood ratio with $h(0, x_0) = p(0, x_0; T, v)$, the correction term $\Psi_T(X) = \lim_{t \uparrow T} \exp\{\int_0^t G(s, X_s)ds\}$, and

$$G(s, x) = [b(s, x) - (\tilde{B}x + \tilde{\beta})]\tilde{r}(s, x) - \frac{1}{2}[a(s, x) - \tilde{a}][l(s)^2 M(s) - \tilde{r}(s, x)^2]$$

where $\tilde{r}(s, x) = l(s)M(s)[v - \mu(s) - l(s)x]$.

To establish Theorem 5.1, we use the auxiliary results proved by Corstanje et al. (2023, Theorems 5.9-5.11). In their diffusion setting, the authors introduce the Lyapunov function

$$V(t, x) = \frac{H(t, x)}{\log(\frac{1}{T-t})}$$

with $H(t, x)$ as defined in (5.5), to construct suitable localising events and control the singular behaviour near the terminal time. In particular, Corstanje et al. (2023, Theorem 5.10) establish the bounds required to control the likelihood ratio terms on these events, while Corstanje et al. (2023, Theorem 5.11) establish the almost sure boundedness needed to show that the terminal localisation event has full probability under the guided law. Together,

with Assumptions 5.1-5.4 and Corstanje et al. (2023, Theorem 5.9), the results of Corstanje et al. (2023) provide the estimates needed to verify the terminal time conditions introduced in Chapter 4 and can therefore apply Theorem 4.1 of this dissertation to obtain the equivalence of the conditioned and guided laws at T .

The following section examines the choice of the Lyapunov function in more detail and explains why the logarithmic normalisation is particularly well suited to controlling the singular terms as $t \uparrow T$.

5.3 The Lyapunov Function

Lyapunov methods provide a way of controlling the behaviour of a stochastic process without requiring a closed form solution of the underlying SDE. As discussed by Mao (2007, pp. 140-185), one introduces a non-negative function V and uses Itô's formula and the infinitesimal generator to study the behaviour of $V(t, X_t)$. In the present setting, however, V is not used to study asymptotic stability as $t \rightarrow \infty$. Instead, it is used as a localisation function to control the guided process near the finite terminal time T , where the guiding term becomes singular.

Following Corstanje et al. (2023), we choose V so that the boundedness of $V(t, X_t^\circ)$ defines the localising events

$$A_k(u) = \left\{ \sup_{t_0 \leq t < u} V(t, X_t^\circ) \leq k \right\}$$

on which the terms of the likelihood ratio are controlled. The Lyapunov method is then used to show that the terminal localisation event $A(T)$ has full probability under the guided law, allowing the localisation to be removed when extending the change of measure to T . Corstanje et al. (2023) use the Lyapunov function

$$V(t, x) = \frac{H(t, x)}{\log(\frac{1}{T-t})}.$$

in establishing terminal time equivalence, although the role of this particular normalisation is not discussed in detail. To examine its role, we compare it with the polynomial family

$$V_\beta(t, x) = (T - t)^\beta H(t, x), \quad 0 < \beta < 1.$$

and show why logarithmic control is more suitable in this setting.

On the event $A_k(u)$,

$$V_\beta(t, X_t^\circ) \leq k, \quad t_0 \leq t < u,$$

and hence

$$H(t, X_t^\circ) \leq k(T - t)^{-\beta}.$$

Thus, this choice permits $H(t, X_t^\circ)$ to grow polynomially as $t \uparrow T$. The question is whether this growth remains compatible with the singular terms appearing in the likelihood ratio estimates.

Following the proof of Corstanje et al. (2023, Theorem 5.11, (A7)). Their proof obtains bounds that contain terms of the form

$$(T - t)^{-1/2} H(t, X_t^\circ)^{1/2}, \quad (T - t)^{-1/2} H(t, X_t^\circ)^{3/2}, \quad (T - t)^{\alpha-1} H(t, X_t^\circ).$$

Using

$$H(t, X_t^\circ) \leq k(T - t)^{-\beta},$$

the first term becomes

$$(T - t)^{-1/2} H(t, X_t^\circ)^{1/2} \leq \sqrt{k}(T - t)^{\frac{-(1+\beta)}{2}}.$$

For this bound to be integrable near T , its exponent must be greater than -1 . Hence,

$$-\frac{1+\beta}{2} > -1$$

or equivalently, $\beta < 1$. Similarly, the second term becomes

$$(T - t)^{-1/2} H(t, X_t^\circ)^{3/2} \leq k^{3/2}(T - t)^{\frac{-(1+3\beta)}{2}}$$

where integrability requires

$$-\frac{1+3\beta}{2} > -1$$

which gives

$$\frac{1+3\beta}{2} < 1 \iff \beta < \frac{1}{3}.$$

Finally, the third term becomes

$$(T - t)^{\alpha-1} H(t, x) \leq k(T - t)^{\alpha-1-\beta}$$

where in order to have integrability, we would need

$$\alpha - 1 - \beta > -1 \iff \beta < \alpha.$$

Combining these conditions gives

$$\beta < \min \left\{ \frac{1}{3}, \alpha \right\}.$$

Although we initially considered $0 < \beta < 1$, the likelihood ratio estimates therefore impose a stronger restriction $\beta < \frac{1}{3}$ together with $\beta < \alpha$. Since Assumption 5.4 only requires $\alpha > 0$, therefore one can choose $\beta > 0$ sufficiently small so that $\beta < \min\{\frac{1}{3}, \alpha\}$. However, there is no single positive β that satisfies this condition uniformly for all $\alpha > 0$. Consequently, a polynomial Lyapunov function cannot be used uniformly under the assumptions of the theorem.

By contrast, for the logarithmic choice of Corstanje et al. (2023),

$$V(t, x) = \frac{H(t, x)}{\log(\frac{1}{T-t})}$$

shows that boundedness of V on $A_k(t)$ implies

$$H(t, X_t^\circ) \leq k \log \left(\frac{1}{T-t} \right).$$

Hence, $H(t, X_t^\circ)$ is allowed to grow only logarithmically as $t \uparrow T$. Since logarithmic growth is slower, this introduces less additional growth into the singular likelihood ratio terms and remains compatible with the estimates for every $\alpha > 0$.

The logarithmic Lyapunov function is also compatible with the stochastic estimates used in their proof. Corstanje et al. (2023, Lemma A.2) obtain a logarithmic almost sure upper bound for the martingale, which is subsequently used in proving the almost sure boundedness result of Corstanje et al. (2023, Theorem 5.11). Thus, the logarithmic Lyapunov function is well suited both to the integrability requirements of the likelihood ratio terms and to the pathwise martingale estimates near the terminal time.

Therefore, the choice

$$V(t, x) = \frac{H(t, x)}{\log(\frac{1}{T-t})}$$

provides sufficient control of the growth of H near T while remaining compatible with the terminal time estimates to have equivalence of the guided and conditioned laws.

5.4 Connection with Neural Guided Diffusion Bridges

The previous sections considered a classical guided proposal obtained by replacing the intractable guiding term of the true conditioned process with one derived from an auxiliary diffusion. Although this construction yields a tractable process satisfying the required terminal constraint, the resulting behaviour may still differ from that of the true bridge when the auxiliary process is a poor approximation to the original diffusion. One recent approach to reducing this discrepancy is the neural guided diffusion bridge proposed by Yang et al. (2025).

If we consider the true bridge to be:

$$dX_t^* = [b(t, X_t^*) + a(t, X_t^*)r(t, X_t^*)]dt + \sigma(t, X_t^*)dW_t^*$$

and the guided process is:

$$dX_t^\circ = [b(t, X_t^\circ) + a(t, X_t^\circ)\tilde{r}(t, X_t^\circ)]dt + \sigma(t, X_t^\circ)dW_t^\circ.$$

The difference between the two drift coefficients is therefore $a(t, x)(r(t, x) - \tilde{r}(t, x))$. Thus, the accuracy of the guided process depends on how well the auxiliary quantity $\tilde{r}$ approximates the intractable quantity r .

The neural guided diffusion bridge will consist of the guided proposal with an additional correction drift term that is parameterised by a learnable neural network. Its SDE is:

$$dX_t^\bullet = [b(t, X_t^\bullet) + a(t, X_t^\bullet)\tilde{r}(t, X_t^\bullet) + \sigma(t, X_t^\bullet)v_\theta(t, X_t^\bullet)]dt + \sigma(t, X_t^\bullet)dW_t^\bullet$$

where $v_\theta : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d, \theta \in \Theta$ and Θ is the parameter space. The additional drift term is therefore σv_θ . The aim is not to replace the guided proposal, but rather to learn the part of the true bridge drift that is not already captured by the auxiliary process.

To identify the role of the neural correction, we compare the drift of the neural guided process directly with that of the true bridge. Hence, the remaining discrepancy from the true bridge is:

$$\begin{aligned} b^* - b_\theta^\bullet &= (b + ar) - (b + a\tilde{r} + \sigma v_\theta) = a(r - \tilde{r}) - \sigma v_\theta \\ &= \sigma \sigma^\top (r - \tilde{r}) - \sigma v_\theta \\ &= \sigma [\sigma^\top (r - \tilde{r}) - v_\theta]. \end{aligned}$$

Thus, if the neural network learns $v_\theta = \sigma^\top(r - \tilde{r})$ then the discrepancy between the neural guided process and the true bridge vanishes and the neural guided drift would be the drift of the true conditioned process i.e.

$$b + a\tilde{r} + \sigma v_\theta = b + a\tilde{r} + \sigma\sigma^\top(r - \tilde{r}) = b + a\tilde{r} + a(r - \tilde{r}) = b + ar.$$

Consequently, the optimal neural correction is

$$v_{opt}(t, x) = \sigma(t, x)^\top(r(t, x) - \tilde{r}(t, x)).$$

In this sense, the neural network learns the error in the classical guiding term expressed in the Brownian, or noise, coordinates of the SDE.

In order to learn v_θ , Yang et al. (2025) introduce an objective in which the family of measures $\{\mathbb{P}_\theta^\bullet : \theta \in \Theta\}$ approximates $\mathbb{P}^*$ via a variational class. The aim is to choose θ so that the law of the neural guided process is as close as possible to the true bridge law, with this discrepancy measured by the Kullback-Leibler divergence. Yang et al. (2025, Theorem 4), Theorem 5.2 below, shows that minimising $\theta \mapsto D_{KL}(\mathbb{P}_\theta^\bullet \parallel \mathbb{P}^*)$ is equivalent to minimising the loss $L(\theta)$ as defined below.

Following Yang et al. (2025), assume that v_θ is bounded and Lipschitz continuous in x , uniformly in t . Boundedness ensures that Novikov's condition is satisfied, which justifies the Girsanov change of measure from $\mathbb{P}^\circ$ to $\mathbb{P}_\theta^\bullet$ used in the proof below.

The following result is due to Yang et al. (2025, Theorem 4). We reproduce the derivation and work through the proof to show how the Girsanov change of measure and the guided proposal likelihood ratio combine to produce the stated variational loss.

Theorem 5.2. *If we define the loss function by*

$$L(\theta) := \mathbb{E}_\theta^\bullet \int_0^T \left\{ \frac{1}{2} \|v_\theta(s, X_s)\|^2 - G(s, X_s) \right\} ds$$

where

$$G(s, x) = \langle b(s, x) - \tilde{b}(s, x), \tilde{r}(s, x) \rangle - \frac{1}{2} \text{tr}([a(s, x) - \tilde{a}(s)][\tilde{H}(s) - (\tilde{r}\tilde{r}^\top)(s, x)]),$$

$\tilde{H}(s) = L^\top(s)M(s)L(s)$, and $\tilde{r}(s, x)$ as defined in Section 5.1. Then,

$$D_{KL}(\mathbb{P}_\theta^\bullet \parallel \mathbb{P}^*) = L(\theta) - \log \frac{\tilde{h}(0, x_0)}{h(0, x_0)}.$$

Proof:

By definition of the KL divergence:

$$\begin{aligned}
D_{KL}(\mathbb{P}_\theta^\bullet \parallel \mathbb{P}^*) &= \mathbb{E}_\theta^\bullet \left[\log \frac{d\mathbb{P}_\theta^\bullet}{d\mathbb{P}^*} \right] \\
&= \mathbb{E}_\theta^\bullet \left[\log \left(\frac{d\mathbb{P}_\theta^\bullet}{d\mathbb{P}^\circ} \frac{d\mathbb{P}^\circ}{d\mathbb{P}^*}(X) \right) \right] \\
&= \mathbb{E}_\theta^\bullet \left[\log \left(\frac{d\mathbb{P}_\theta^\bullet}{d\mathbb{P}^\circ}(X) \right) \right] - \mathbb{E}_\theta^\bullet \left[\log \left(\frac{d\mathbb{P}^*}{d\mathbb{P}^\circ}(X) \right) \right].
\end{aligned}$$

For the first term, $\mathbb{E}_\theta^\bullet[\log(\frac{d\mathbb{P}_\theta^\bullet}{d\mathbb{P}^\circ}(X))]$, we will define

$$\varkappa_t := \exp \left(\int_0^t v_\theta^\top(s, X_s) dW_s^\circ - \frac{1}{2} \int_0^t \|v_\theta(s, X_s)\|^2 ds \right).$$

By the boundedness assumption on v_θ , Novikov's condition holds and $\varkappa_t$ is a martingale. We may therefore define

$$d\mathbb{P}_\theta^\bullet = \varkappa_t d\mathbb{P}^\circ.$$

Therefore,

$$\log \frac{d\mathbb{P}_\theta^\bullet}{d\mathbb{P}^\circ} = \int_0^T v_\theta^\top(s, X_s) dW_s^\circ - \frac{1}{2} \int_0^T \|v_\theta(s, X_s)\|^2 ds$$

Using Girsanov's theorem, as defined in Section 2.4, under the new measure $\mathbb{P}_\theta^\bullet$, we have: $W_t^\bullet = W_t^\circ - \int_0^t v_\theta(s, X_s) ds$, rearranging this term gives:

$W_t^\circ = W_t^\bullet + \int_0^t v_\theta(s, X_s) ds$ which is $dW_t^\circ = dW_t^\bullet + v_\theta(t, X_t) dt$ in the differential form. Hence,

$$\begin{aligned}
\log \frac{d\mathbb{P}_\theta^\bullet}{d\mathbb{P}^\circ} &= \int_0^T v_\theta^\top(s, X_s) (dW_s^\bullet + v_\theta(s, X_s) ds) - \frac{1}{2} \int_0^T \|v_\theta(s, X_s)\|^2 ds \\
&= \int_0^T v_\theta^\top(s, X_s) dW_s^\bullet + \int_0^T v_\theta^\top(s, X_s) v_\theta(s, X_s) ds - \frac{1}{2} \int_0^T \|v_\theta(s, X_s)\|^2 ds \\
&= \int_0^T v_\theta^\top(s, X_s) dW_s^\bullet + \frac{1}{2} \int_0^T \|v_\theta(s, X_s)\|^2 ds.
\end{aligned}$$

Taking expectation on both sides:

$$\begin{aligned}
\mathbb{E}_\theta^\bullet \left[\log \frac{d\mathbb{P}_\theta^\bullet}{d\mathbb{P}^\circ} \right] &= \mathbb{E}_\theta^\bullet \left[\int_0^T v_\theta^\top(s, X_s) dW_s^\bullet + \frac{1}{2} \int_0^T \|v_\theta(s, X_s)\|^2 ds \right] \\
&= \mathbb{E}_\theta^\bullet \left[\frac{1}{2} \int_0^T \|v_\theta(s, X_s)\|^2 ds \right]
\end{aligned}$$

as the expectation of the stochastic integral is zero i.e. $\mathbb{E}_\theta^\bullet[\int_0^T v_\theta^\top(s, X_s) dW_s^\bullet] = 0$.

For the second term, from Yang et al. (2025, Theorem 2) the Radon-Nikodym derivative would be:

$$\frac{d\mathbb{P}^*}{d\mathbb{P}^\circ} = \frac{\tilde{h}(0, x_0)}{h(0, x_0)} \Psi_T(X)$$

where $\Psi_T(X) = \exp\{\int_0^T \frac{(\partial_t + \mathcal{A})\tilde{h}}{\tilde{h}}(s, X_s) ds\}$. Taking log and expectation on both sides:

$$\mathbb{E}_\theta^\bullet \left[\log \frac{d\mathbb{P}^*}{d\mathbb{P}^\circ} \right] = \log \frac{\tilde{h}(0, x_0)}{h(0, x_0)} + \mathbb{E}_\theta^\bullet \left[\int_0^T G(t, X_s) ds \right].$$

Thus,

$$\begin{aligned} D_{KL}(\mathbb{P}_\theta^\bullet \parallel \mathbb{P}^*) &= \mathbb{E}_\theta^\bullet \left[\frac{1}{2} \int_0^T \|v_\theta\|^2 ds \right] - \log \frac{\tilde{h}(0, x_0)}{h(0, x_0)} - \mathbb{E}_\theta^\bullet \left[\int_0^T G(t, X_s) ds \right] \\ &= \mathbb{E}_\theta^\bullet \left[\frac{1}{2} \int_0^T \|v_\theta\|^2 ds - \int_0^T G(t, X_s) ds \right] - \log \frac{\tilde{h}(0, x_0)}{h(0, x_0)}. \blacksquare \end{aligned}$$

Yang et al. (2025, Section 5.1) consider the following one-dimensional Brownian motion example and identify the corresponding optimal correction. However, they do not work through the example so we develop it further by deriving the variational loss as a function of θ since the law of $X^\bullet$ itself depends on θ , and although v_θ appears in the quadratic term of the loss, the dependence of $G(s, X_s)$ on θ is indirect. We will show that minimising this loss recovers the optimal correction.

Consider the diffusion for a one-dimensional Brownian motion, with constant drift and diffusion coefficients

$$dX_t = \gamma dt + \sigma dW_t$$

and as the transition density of Brownian motion is Gaussian, the fully observed process conditioned on $\{X_T = v\}$ has the SDE

$$dX_t^* = \frac{v - X_t^*}{T - t} dt + \sigma dW_t^*$$

with the auxiliary process

$$\tilde{X}_t = \sigma W_t$$

so the guided neural network solves the SDE

$$dX_t^\bullet = \left\{ \gamma + \frac{v - X_t^\bullet}{T - t} + \sigma v_\theta(t, X_t^\bullet) \right\} dt + \sigma dW_t \quad (5.7)$$

For our example, we will consider $v_\theta(t, x) = \theta$. We know that

$$G(t, x) = (b(t, x) - \tilde{b}(t, x))\tilde{r}(t, x) - \frac{1}{2}(a(t, x) - \tilde{a}(t, x))(\tilde{H}(t) - \tilde{r}(t, x)^2)$$

where $\tilde{r}(t, x) = \nabla_x \log \tilde{p}(T, v \mid t, x) = \frac{v-x}{\sigma^2(T-t)}$, $b(t, x) = \gamma$, $\tilde{b}(t, x) = 0$, $a(t, x) = \tilde{a}(t, x) = \sigma^2$. Hence,

$$G(t, x) = \frac{\gamma(v-x)}{\sigma^2(T-t)}.$$

Our loss function becomes

$$\begin{aligned} L(\theta) &= \mathbb{E}_\theta^\bullet \left[\int_0^T \left\{ \frac{1}{2} \|v_\theta(t, X_t)\|^2 - G(t, X_t) \right\} dt \right] \\ &= \mathbb{E}_\theta^\bullet \left[\int_0^T \left\{ \frac{1}{2}\theta^2 - \frac{\gamma(v - X_t^\bullet)}{\sigma^2(T-t)} \right\} dt \right] \\ &\stackrel{\text{Fubini}}{=} \int_0^T \mathbb{E}_\theta^\bullet \left[\frac{1}{2}\theta^2 - \frac{\gamma(v - X_t^\bullet)}{\sigma^2(T-t)} \right] dt \\ &= \int_0^T \left[\frac{1}{2}\theta^2 - \frac{\gamma(v - \mathbb{E}_\theta^\bullet X_t^\bullet)}{\sigma^2(T-t)} \right] dt \end{aligned}$$

To apply Fubini's theorem, it is sufficient to verify

$$\mathbb{E}_\theta^\bullet \left[\int_0^T \left| \frac{1}{2}\theta^2 - \frac{\gamma(v - X_t^\bullet)}{\sigma^2(T-t)} \right| dt \right] < \infty$$

for $0 \leq t < T$. Since $\int_0^T \frac{1}{2}\theta^2 dt = \frac{1}{2}\theta^2 T < \infty$, it remains to show

$$\int_0^T \mathbb{E}_\theta^\bullet \left[\left| \frac{v - X_t^\bullet}{T-t} \right| \right] dt < \infty.$$

Let $Y_t = \frac{v - X_t^\bullet}{T-t} = (T-t)^{-1}(v - X_t^\bullet)$. Itô's lemma says for a process

$$dX_t = b_t dt + \sigma_t dW_t$$

and a function $u(t, x)$,

$$du(t, X_t) = \partial_t u(t, X_t) dt + \partial_x u(t, X_t) dX_t + \frac{1}{2} \partial_{xx} u(t, X_t) \sigma_t^2 dt.$$

Let $u(t, x) = \frac{v-x}{T-t}$ so that $Y_t = u(t, X_t^\bullet)$. For our process, $\sigma_t = \sigma$,

$$\partial_t u(t, x) = \frac{v-x}{(T-t)^2}, \quad \partial_x u(t, x) = -\frac{1}{T-t}, \quad \partial_{xx} u(t, x) = 0.$$

Hence,

$$\begin{aligned}
dY_t &= \frac{v - X_t^\bullet}{(T-t)^2} dt - \frac{1}{T-t} dX_t^\bullet + \frac{1}{2}(0)\sigma^2 dt \\
&= \frac{-1}{T-t} dX_t^\bullet + \frac{v - X_t^\bullet}{(T-t)^2} dt \\
&\stackrel{(5.7)}{=} \frac{-1}{T-t} \left[\left(\gamma + \frac{v - X_t^\bullet}{T-t} + \sigma\theta \right) dt + \sigma dW_t \right] + \frac{v - X_t^\bullet}{(T-t)^2} dt \\
&= -\frac{\gamma + \sigma\theta}{T-t} dt - \frac{v - X_t^\bullet}{(T-t)^2} dt - \frac{\sigma}{T-t} dW_t + \frac{v - X_t^\bullet}{(T-t)^2} dt \\
&= -\frac{\gamma + \sigma\theta}{T-t} dt - \frac{\sigma}{T-t} dW_t
\end{aligned}$$

Integrating both sides

$$Y_t = \frac{v - x_0}{T} - (\gamma + \sigma\theta) \log \frac{T}{T-t} - \sigma \int_0^t \frac{1}{T-s} dW_s \quad (5.8)$$

Therefore,

$$\mathbb{E}_\theta^\bullet |Y_t| \leq \left| \frac{v - x_0}{T} - (\gamma + \sigma\theta) \log \frac{T}{T-t} \right| + \sigma \sqrt{\frac{1}{T-t} - \frac{1}{T}} \quad (5.9)$$

where the inequality of (5.9) follows from a Cauchy-Schwartz argument:

$$\mathbb{E}_\theta^\bullet \left| \int_0^t \frac{1}{T-s} dW_s^\bullet \right| \leq \sqrt{\mathbb{E}_\theta^\bullet \left[\left(\int_0^t \frac{1}{T-s} dW_s^\bullet \right)^2 \right]}$$

and using Itô's isometry

$$\mathbb{E}_\theta^\bullet \left[\left(\int_0^t \frac{1}{T-s} dW_s^\bullet \right)^2 \right] = \int_0^t \frac{1}{(T-s)^2} ds = \frac{1}{T-t} - \frac{1}{T}.$$

Therefore,

$$\mathbb{E}_\theta^\bullet \left| \int_0^t \frac{1}{T-s} dW_s^\bullet \right| \leq \sqrt{\frac{1}{T-t} - \frac{1}{T}}.$$

As,

$$\int_0^T \log \frac{T}{T-t} dt = T < \infty$$

and

$$\int_0^T (T-t)^{-1/2} dt = 2\sqrt{T} < \infty$$

then $\int \mathbb{E}_\theta^\bullet |Y_t| < \infty$, justifying the use of Fubini.

Going back to our loss function, it remains to calculate $\mathbb{E}_\theta^\bullet X_t^\bullet$.

Taking expectations on both sides from (5.8):

$$\mathbb{E}_\theta^\bullet(Y_t) = \frac{v - x_0}{T} - (\gamma + \sigma\theta) \log \frac{T}{T-t}$$

where $\mathbb{E}_\theta^\bullet(\sigma \int_0^t \frac{1}{T-s} dW_s) = 0$ as the expectation of the stochastic integral is zero.

From the definition of Y_t : $Y_t = \frac{v - X_t^\bullet}{T-t} \implies X_t^\bullet = v - (T-t)Y_t$.

Taking expectation on both sides

$$\begin{aligned} \mathbb{E}_\theta^\bullet(X_t^\bullet) &= v - (T-t)\mathbb{E}_\theta^\bullet(Y_t) \\ &= v - \frac{(T-t)}{T}(v - x_0) + (\gamma + \sigma\theta)(T-t) \log \left(\frac{T}{T-t} \right) \end{aligned}$$

Our loss function becomes:

$$\begin{aligned} L(\theta) &= \int_0^T \left[\frac{1}{2}\theta^2 - \frac{\gamma(v - \mathbb{E}_\theta^\bullet X_t^\bullet)}{\sigma^2(T-t)} \right] dt \\ &= \int_0^T \left[\frac{1}{2}\theta^2 - \frac{\gamma}{\sigma^2} \left(\frac{v - x_0}{T} - (\gamma + \sigma\theta) \log \left(\frac{T}{T-t} \right) \right) \right] dt \\ &\stackrel{(*)}{=} \frac{T}{2}\theta^2 - \frac{\gamma(v - x_0)}{\sigma^2} + \frac{\gamma^2 T}{\sigma^2} + \frac{\gamma\theta T}{\sigma} \\ &= \frac{T}{2}\theta^2 + \frac{\gamma\theta T}{\sigma} + \frac{\gamma^2 T}{\sigma^2} - \frac{\gamma(v - x_0)}{\sigma^2} \end{aligned}$$

where $(*)$ follows from $\int_0^T \log \frac{T}{T-t} dt = T$.

If we were to minimise the loss, we would obtain:

$$L'(\theta) = T\theta + \frac{\gamma T}{\sigma} = 0 \implies \sigma T\theta^* = -\gamma T \implies \theta^* = -\frac{\gamma}{\sigma}$$

matching the optimal correction term in Yang et al. (2025). In this example, the correction $v_{opt} = -\frac{\gamma}{\sigma}$ removes the discrepancy introduced by the constant drift γ , so the neural guided drift coincides with that of the true conditioned process. This shows how the variational formulation of Yang et al. (2025) can recover the drift correction needed to transform a guided proposal into the true bridge.

Chapter 6

Discussion and Conclusion

The main objective of this dissertation was to study the relationship between conditioned diffusion processes and guided proposals, with particular focus on the terminal time. Although guided proposals provide a tractable way of simulating paths towards a specified terminal observation, their law does not in general coincide with that of the true conditioned diffusion. For each $t < T$, the two laws can be related through a change of measure; the question considered in this dissertation is whether this relationship extends to the complete interval $[0, T]$.

The difficulty arises from the singular behaviour of the conditioning as $t \uparrow T$. The conditioning function becomes increasingly concentrated around the terminal endpoint, while the guiding term and the likelihood ratio may become unbounded. Consequently, equivalence on every interval $[0, t]$, $t < T$, does not by itself imply equivalence on $[0, T]$.

The terminal time analysis showed that extending the finite time change of measure relationship to T requires more than pointwise convergence of the likelihood ratio. We also need to control the conditioning functions, correction factor, and localising events, so that the limiting identity continues to hold under the guided law. In the one-dimensional diffusion setting, the terminal time conditions can be checked directly through the auxiliary drift, diffusion coefficients, terminal residual and precision $M(t)$, showing how the singular behaviour arises as $t \uparrow T$.

Localisation plays an important role in the analysis. Rather than requiring global boundedness of the likelihood ratio, paths are restricted to events on which a Lyapunov function remains controlled. In the diffusion setting, the

logarithmic Lyapunov function,

$$V(t, x) = \frac{H(t, x)}{\log(1/(T - t))}$$

is well matched to the singularity rates appearing in the likelihood ratio estimates. The comparison with polynomial Lyapunov functions shows that this choice is not arbitrary as polynomial growth may fail to satisfy the required integrability conditions, whereas logarithmic growth remains compatible with the assumptions.

The final part of the dissertation connected the classical guided proposal framework with neural guided diffusion bridges. The neural correction v_θ is used to learn the discrepancy between the tractable guiding term $\tilde{r}$ and the intractable score r of the true bridge. In the one-dimensional Brownian example, the variational loss was derived as a function of θ and its minimiser was

$$\theta^* = -\frac{\gamma}{\sigma}$$

which agrees with the general correction term.

The analysis in this dissertation was mainly restricted to a one-dimensional, fully observed terminal condition, which allowed the singular behaviour and Lyapunov estimates to be analysed in this simplified setting. A further limitation is that theoretical equivalence does not necessarily imply that the proposal provides a good approximation to the conditioned process, the quality of the guided process still depends strongly on the choice of auxiliary diffusion, and a poor approximation may reduce the effectiveness of the importance weights.

The comparison in Chapter 5 shows that the logarithmic Lyapunov function is closely tied to the singularity behaviour appearing in the likelihood ratio estimates. For other classes of conditioned processes, the behaviour may differ, and an alternative Lyapunov function may be more appropriate. A useful direction for further work would be to develop a procedure for selecting a localisation function from the asymptotic behaviour of the guiding and correction terms, rather than choosing the normalisation on a model by model basis.

Finally, in connecting the classical guided proposal framework with neural guided diffusion bridges, rather than fixing an auxiliary process and accepting the resulting discrepancy, a learned correction can be used to bring the proposal closer to the true bridge. The Brownian example in Chapter 5 was

used to show what the optimal correction is. A next step would be to simulate the classical and neural guided proposals numerically and compare how the learned correction affects quantities such as the variability of the likelihood ratio, and sensitivity to the choice of auxiliary process. This would allow the terminal time change of measure theory developed in the dissertation to be related more directly to the performance of neural guided proposals.

Overall, this dissertation studies the terminal time change of measure for guided diffusion bridges and the conditions required to extend the finite time relationship between the guided and conditioned laws to T . The localisation and Lyapunov techniques make the role of these conditions clearer, while the one-dimensional specialisation shows how they can be examined directly and clarifies the choice of logarithmic Lyapunov function in this setting. The connection with neural guided diffusion bridges further shows how a learned correction can account for the difference between the guided proposal and the true bridge. Together, these results connect the terminal time theory of guided proposals with both classical and learning based approaches to diffusion bridges.

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